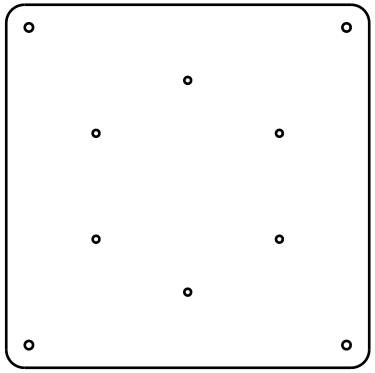
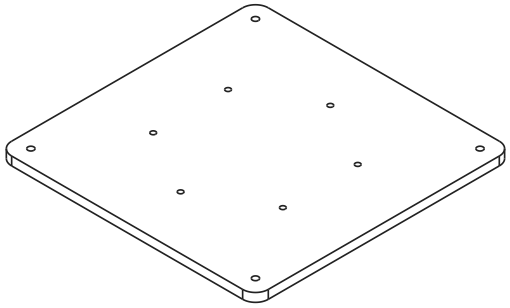


FRONT

LEFT



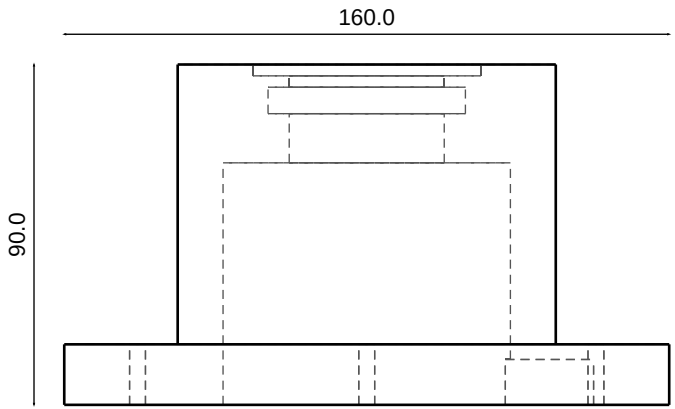
TOP (plan)



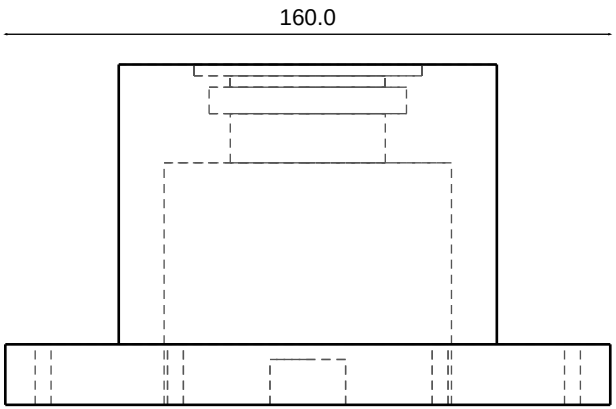
FIRST-ANGLE PROJECTION (ISO 128-30)

ROBOT_BASE_PLATE	
Part no. R01 Make: M	Scale 1:5 A4
Al 6061-T6, 8 mm plate	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 1/50

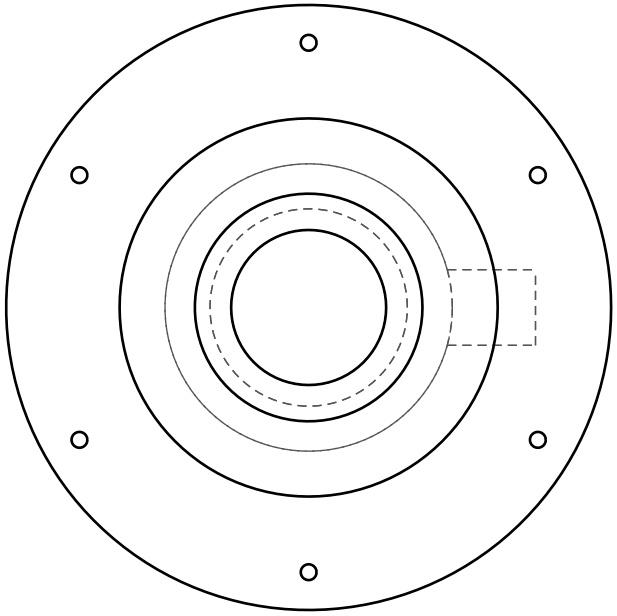
Square base plate, 4x M5 to table, 6x M4 to pedestal. Overall size 240.0 x 240.0 x 8.0 mm. Full dimensions: native SLDPRT / STEP model.



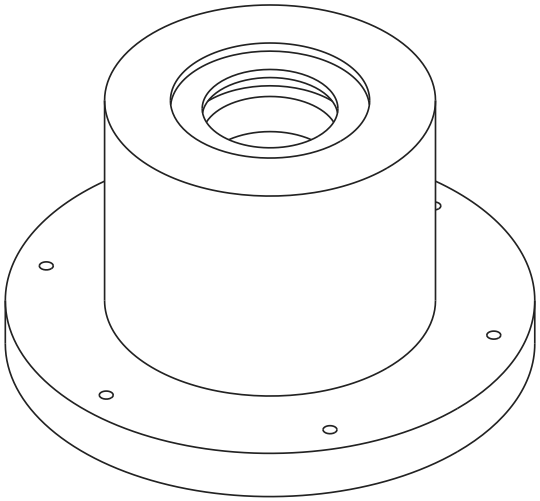
FRONT



LEFT



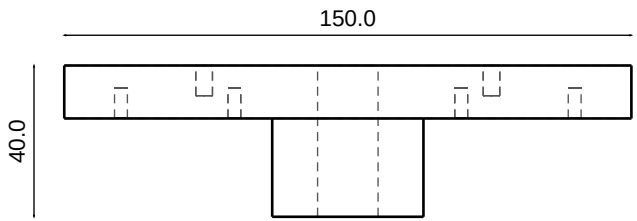
TOP (plan)



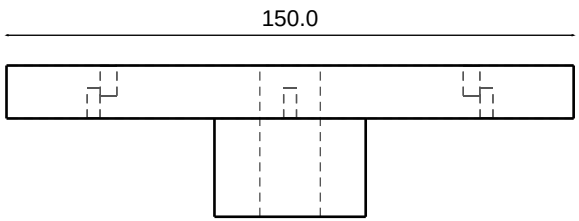
FIRST-ANGLE PROJECTION (ISO 128-30)

ROBOT_BASE	
Part no. R02 Make: P	Scale 1:2 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 2/50

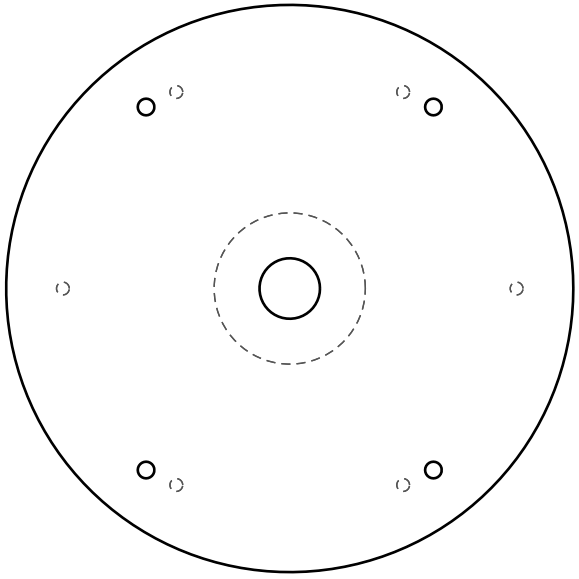
Stepped pedestal; seats 6808 + 51108 J1 bearings. Overall size 160.0 x 160.0 x 90.0 mm. Full dimensions: native SLDPRT / STEP model.



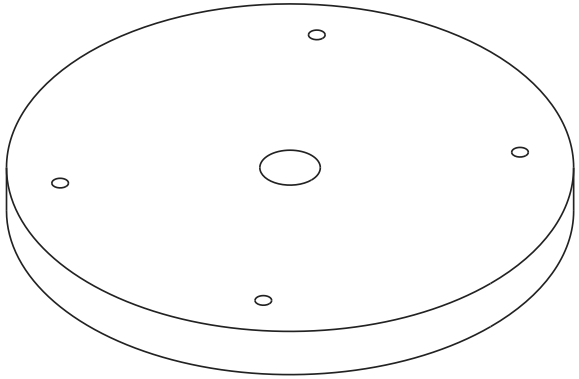
FRONT



LEFT



TOP (plan)



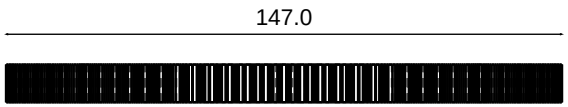
FIRST-ANGLE PROJECTION (ISO 128-30)

ROBOT_TURNTABLE	
Part no. R03 Make: P	Scale 1:2 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 3/50

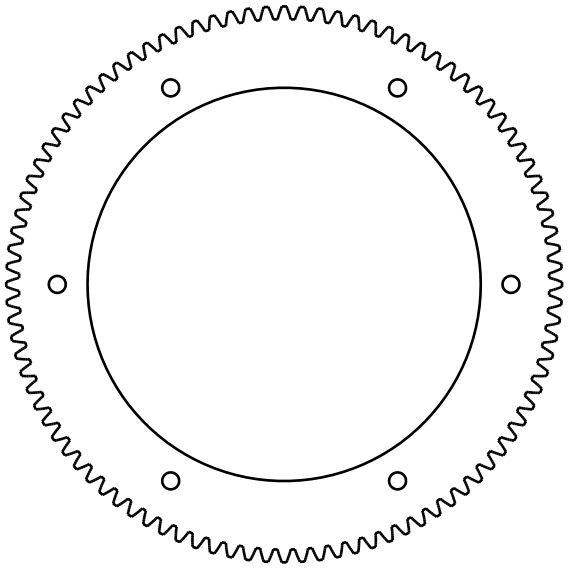
J1 rotating platform with Ø40 spigot. Overall size 150.0 x 150.0 x 40.0 mm. Full dimensions: native SLDPRT / STEP model.



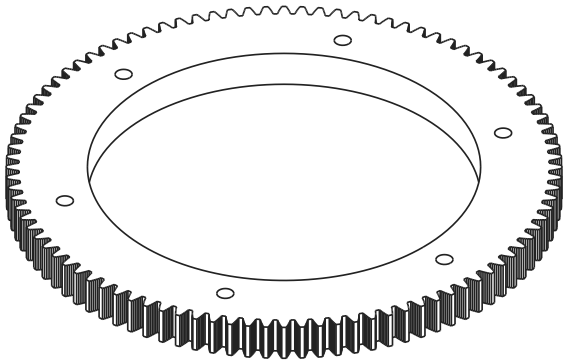
FRONT



LEFT



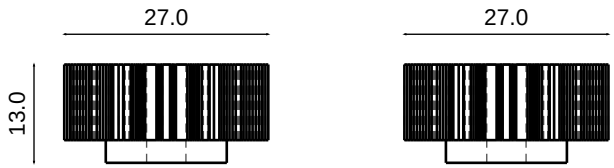
TOP (plan)



FIRST-ANGLE PROJECTION (ISO 128-30)

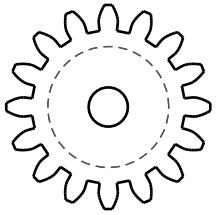
J1_RING_GEAR	
Part no. R04 Make: P	Scale 1:2 A4
PETG (PA-CF preferred)	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 4/50

External spur gear m1.5 z96. Overall size 147.0 x 147.0 x 10.0 mm. Full dimensions: native SLDPRT / STEP model.

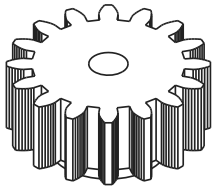


FRONT

LEFT



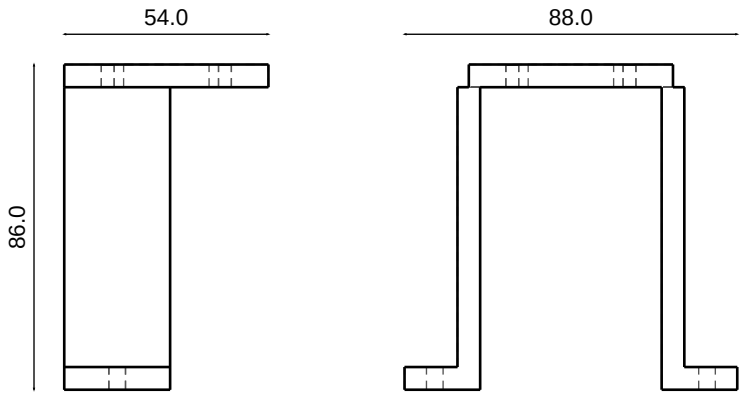
TOP (plan)

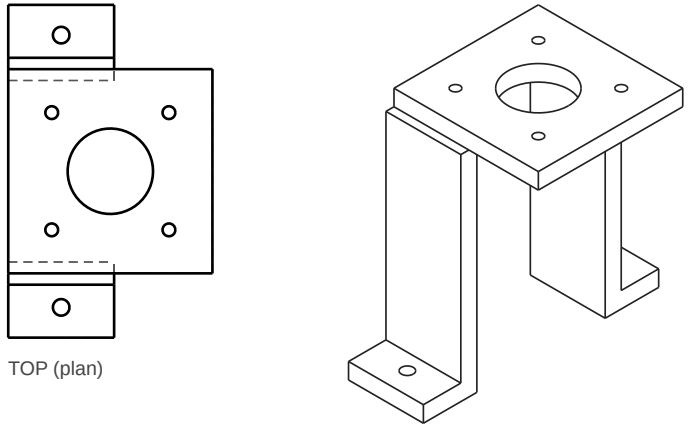


FIRST-ANGLE PROJECTION (ISO 128-30)

J1_PINION	
Part no. R05 Make: P	Scale 1:1 A4
PETG (PA-CF preferred)	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 5/50

Pinion m1.5 z16 on J1 motor (ratio 6:1). Overall size 27.0 x 27.0 x 13.0 mm. Full dimensions: native SLDPRT / STEP model.

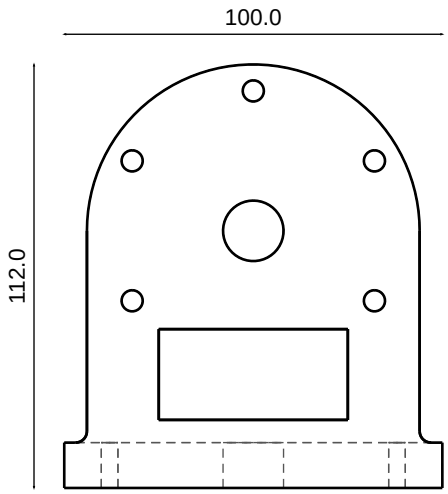




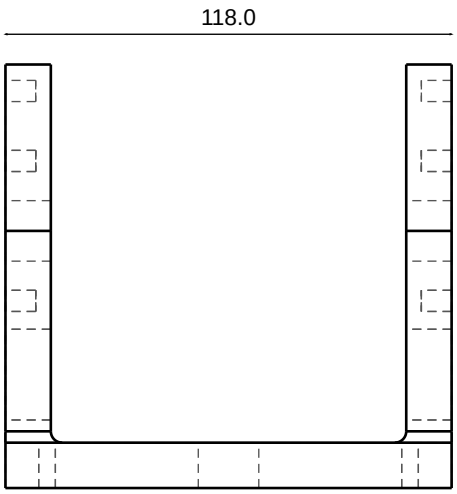
FIRST-ANGLE PROJECTION (ISO 128-30)

J1_MOTOR_BRACKET	
Part no. R06 Make: P	Scale 1:2 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 6/50

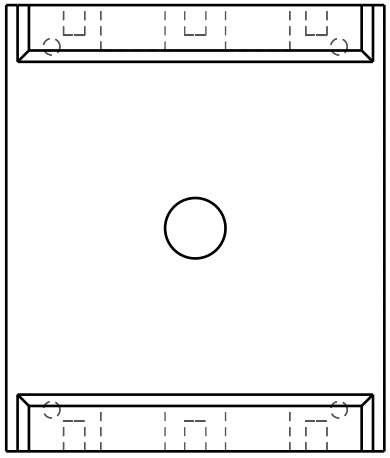
J1 motor bridge bracket. Overall size 54.0 x 88.0 x 86.0 mm. Full dimensions: native SLDPRT / STEP model.



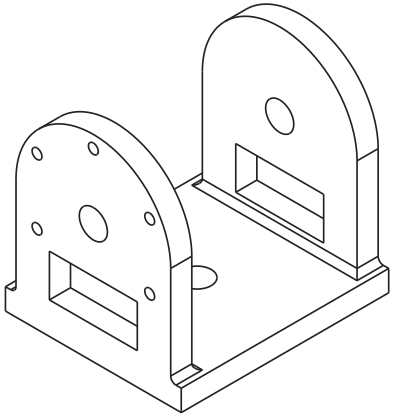
FRONT



LEFT



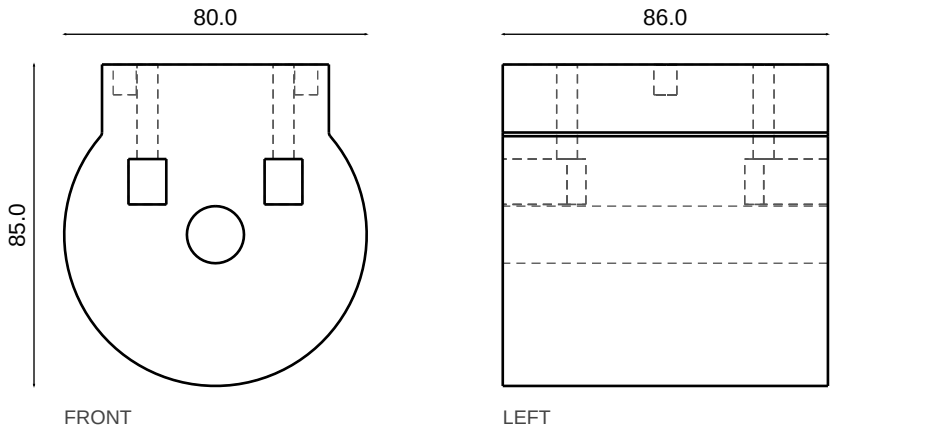
TOP (plan)



FIRST-ANGLE PROJECTION (ISO 128-30)

ROBOT_SHOULDER	
Part no. R07 Make: P	Scale 1:2 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 7/50

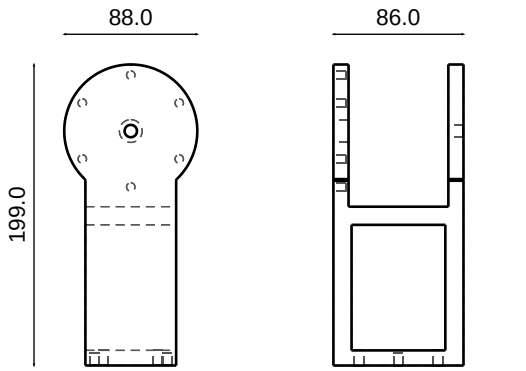
J2 fork; carries both J2 gearboxes. Overall size 100.0 x 118.0 x 112.0 mm. Full dimensions: native SLDPRT / STEP model.



FIRST-ANGLE PROJECTION (ISO 128-30)

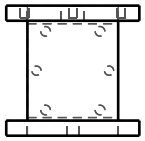
ROBOT_UPPER_ARM_A	
Part no. R09A Make: P	Scale 1:2 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 8/50

Upper arm print half A (J2 hub). Overall size 80.0 x 86.0 x 85.0 mm. Full dimensions: native SLDPRT / STEP model.

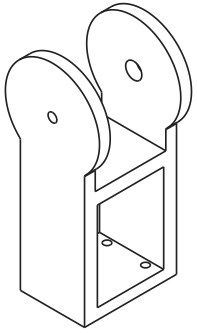


FRONT

LEFT



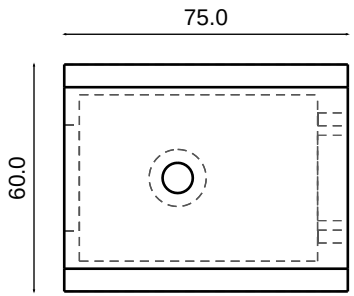
TOP (plan)



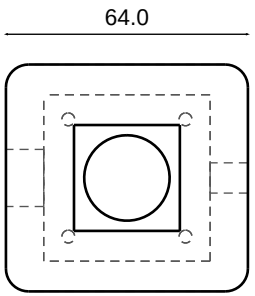
FIRST-ANGLE PROJECTION (ISO 128-30)

ROBOT_UPPER_ARM_B	
Part no. R09B Make: P	Scale 1:5 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 9/50

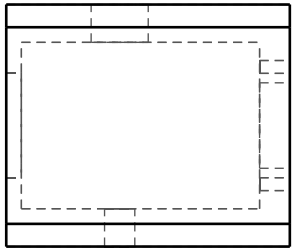
Upper arm print half B (J3 fork). Overall size 88.0 x 86.0 x 199.0 mm. Full dimensions: native SLDPRT / STEP model.



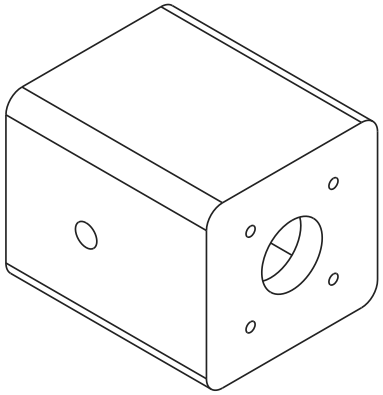
FRONT



LEFT



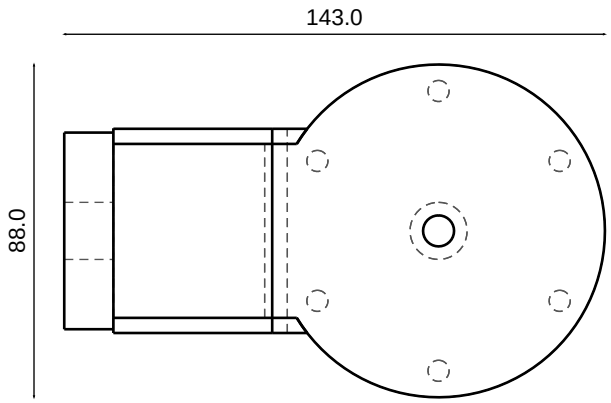
TOP (plan)



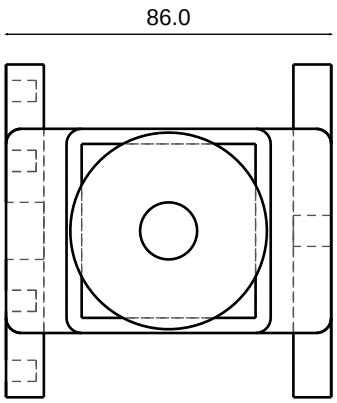
FIRST-ANGLE PROJECTION (ISO 128-30)

ROBOT_ELBOW_HOUSING	
Part no. R10 Make: P	Scale 1:2 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 10/50

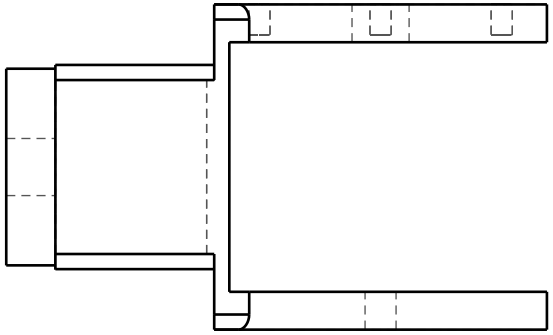
J3 housing, J4 motor inside. Overall size 75.0 x 64.0 x 60.0 mm. Full dimensions: native SLDPRT / STEP model.



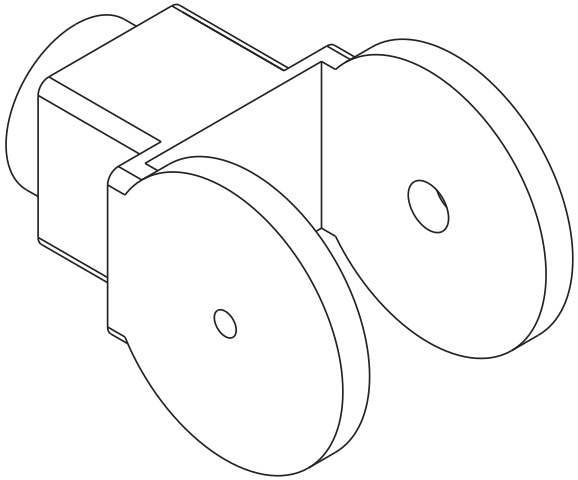
FRONT



LEFT



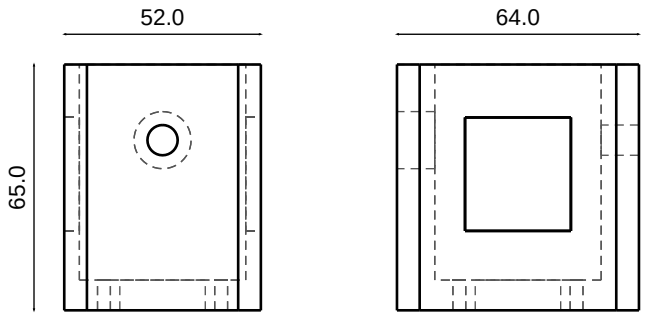
TOP (plan)



FIRST-ANGLE PROJECTION (ISO 128-30)

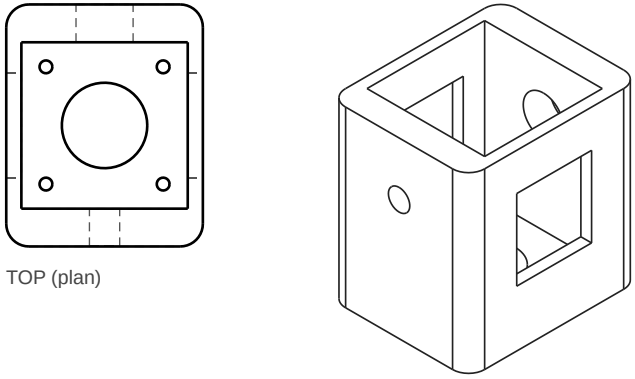
ROBOT_LOWER_ARM	
Part no. R12 Make: P	Scale 1:2 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 11/50

Forearm box section with J5 fork. Overall size 143.0 x 86.0 x 88.0 mm. Full dimensions: native SLDPRT / STEP model.



FRONT

LEFT

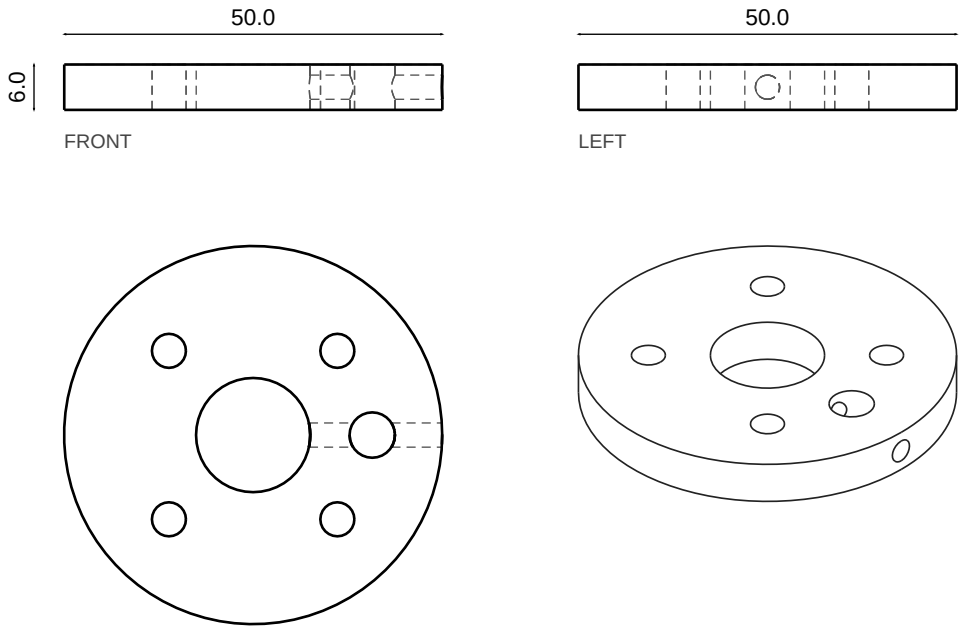


TOP (plan)

FIRST-ANGLE PROJECTION (ISO 128-30)

ROBOT_WRIST	
Part no. R13 Make: P	Scale 1:2 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 12/50

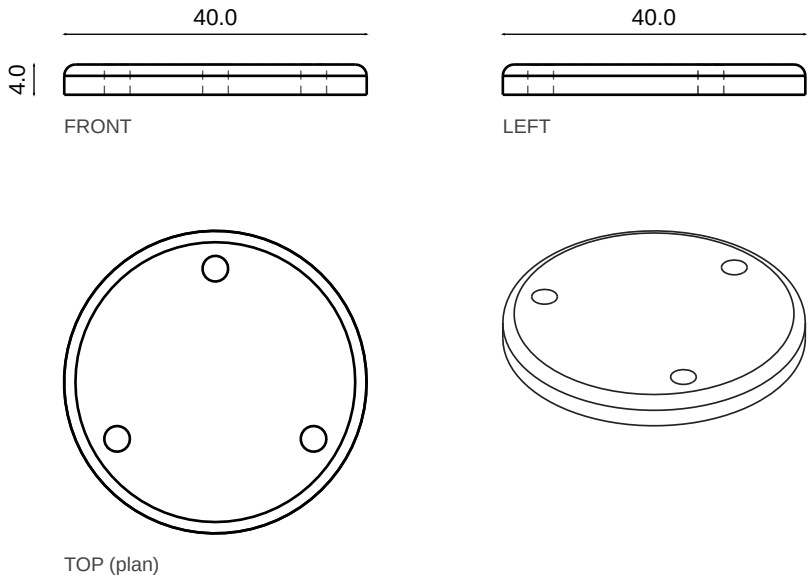
J5 wrist body, J6 motor inside. Overall size 52.0 x 64.0 x 65.0 mm. Full dimensions: native SLDPRT / STEP model.



FIRST-ANGLE PROJECTION (ISO 128-30)

ROBOT_TOOL_FLANGE	
Part no. R14 Make: M	Scale 1:1 A4
Al 6061-T6 (or PETG)	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 13/50

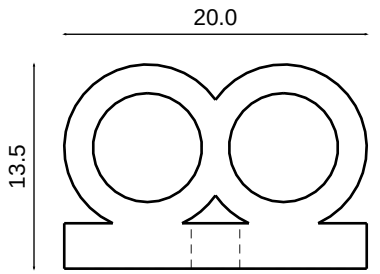
ISO 9409-1-31.5-4-M4 style tool flange. Overall size 50.0 x 50.0 x 6.0 mm. Full dimensions: native SLDPRT / STEP model.



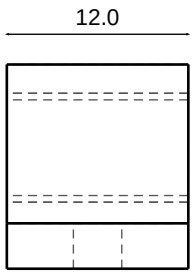
FIRST-ANGLE PROJECTION (ISO 128-30)

JOINT_COVER	
Part no. R15 Make: P	Scale 1:1 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 14/50

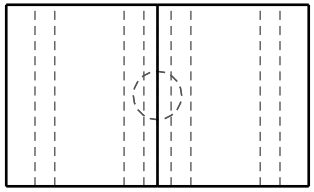
End cap over J3/J5 idler pins. Overall size 40.0 x 40.0 x 4.0 mm. Full dimensions: native SLDPRT / STEP model.



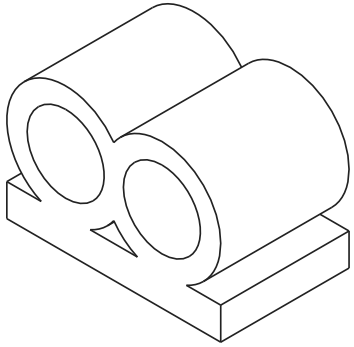
FRONT



LEFT



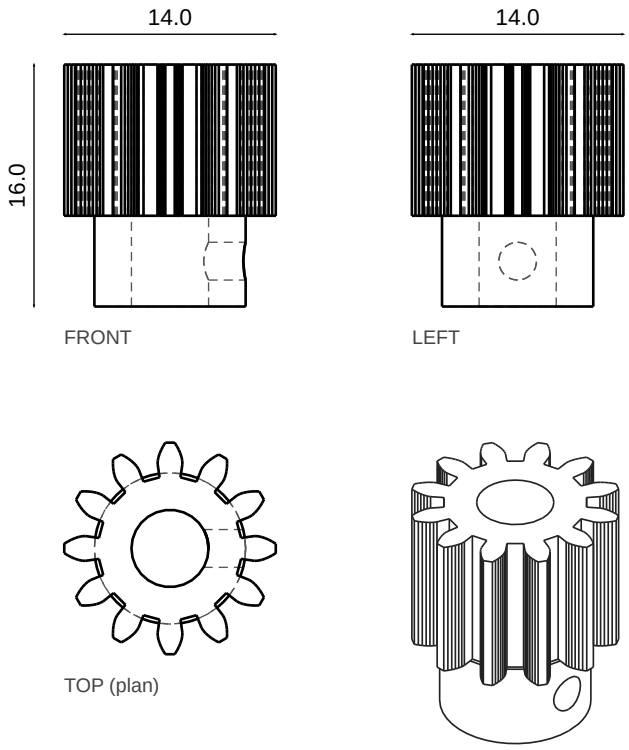
TOP (plan)



FIRST-ANGLE PROJECTION (ISO 128-30)

CABLE_CLAMP	
Part no. R16 Make: P	Scale 2:1 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 15/50

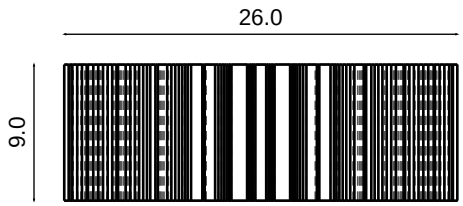
Two-cable clip. Overall size 20.0 x 12.0 x 13.5 mm. Full dimensions: native SLDPRT / STEP model.



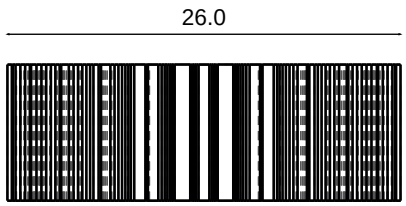
FIRST-ANGLE PROJECTION (ISO 128-30)

GBX_SUN_GEAR	
Part no. G01 Make: P	Scale 2:1 A4
PA-CF or PETG, 100% infill	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 16/50

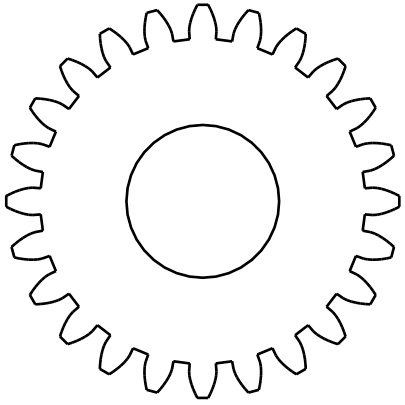
Sun gear m1 z12, D-bore 5. Overall size 14.0 x 14.0 x 16.0 mm. Full dimensions: native SLDPRT / STEP model.



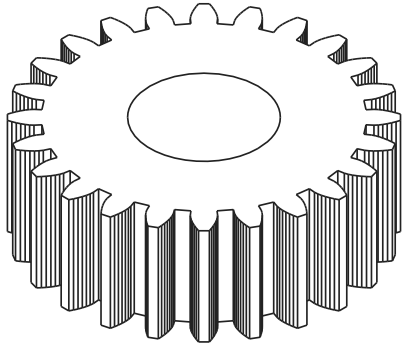
FRONT



LEFT



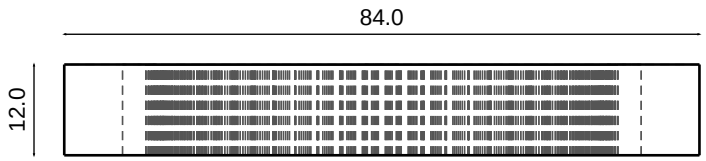
TOP (plan)



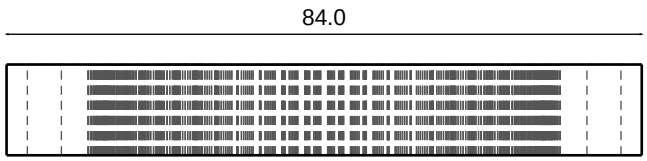
Planet gear m1 z24, MR105 bore. Overall size 26.0 x 26.0 x 9.0 mm. Full dimensions: native SLDPRT / STEP model.

FIRST-ANGLE PROJECTION (ISO 128-30)

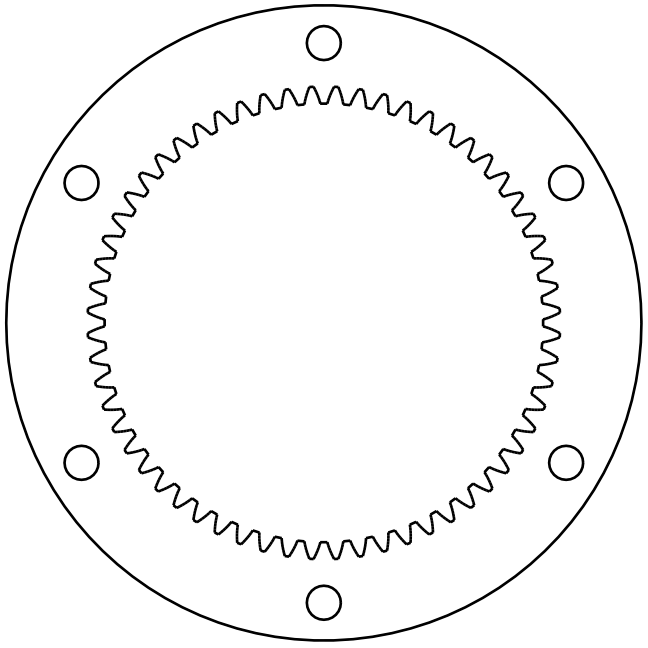
GBX_PLANET_GEAR	
Part no. G02 Make: P	Scale 2:1 A4
PA-CF or PETG, 100% infill	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 17/50



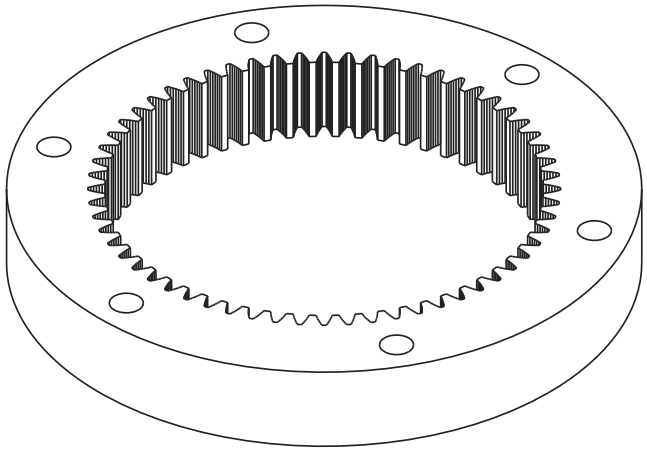
FRONT



LEFT



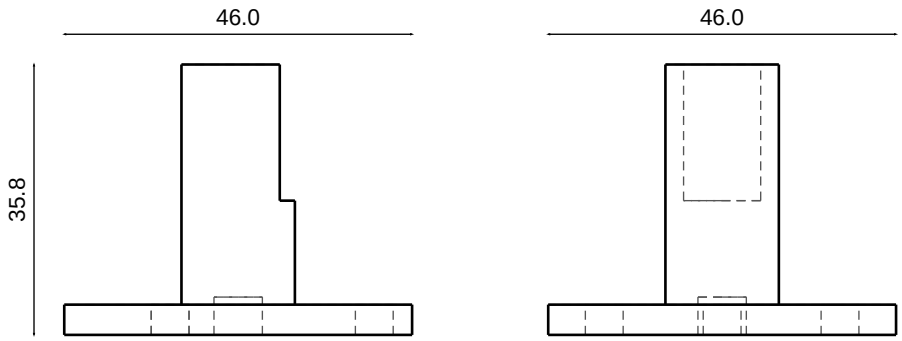
TOP (plan)



FIRST-ANGLE PROJECTION (ISO 128-30)

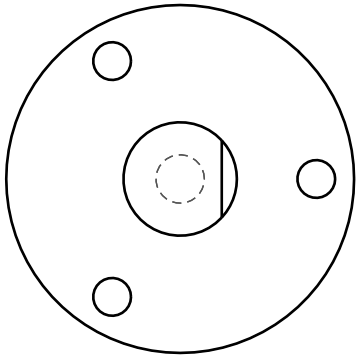
GBX_RING_GEAR	
Part no. G03 Make: P	Scale 1:1 A4
PETG, 100% infill	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 18/50

Internal ring gear m1 z60. Overall size 84.0 x 84.0 x 12.0 mm. Full dimensions: native SLDPRT / STEP model.

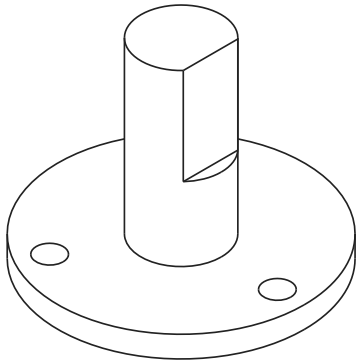


FRONT

LEFT



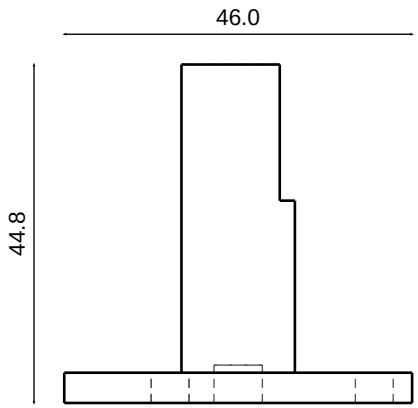
TOP (plan)



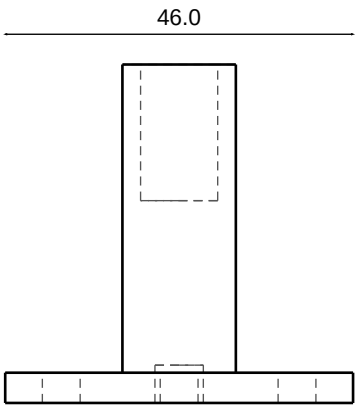
FIRST-ANGLE PROJECTION (ISO 128-30)

GBX_CARRIER_OUTPUT	
Part no. G04 Make: P	Scale 1:1 A4
PA-CF or PETG, 100% infill (steel	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 19/50

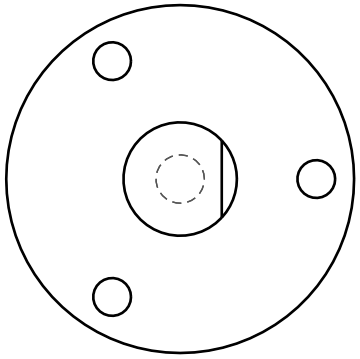
Planet carrier + Ø15 output shaft, config STD. Overall size 46.0 x 46.0 x 35.8 mm. Full dimensions: native SLDPRT / STEP model.



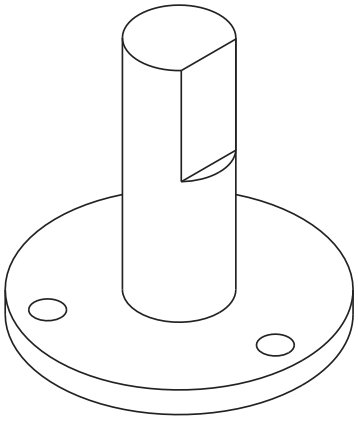
FRONT



LEFT



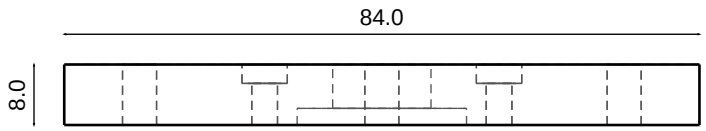
TOP (plan)



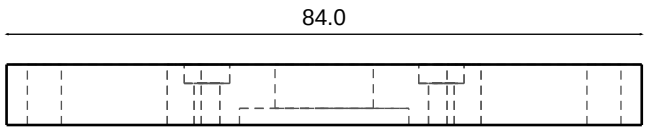
FIRST-ANGLE PROJECTION (ISO 128-30)

GBX_CARRIER_OUTPUT_J2_LONG	
Part no. G04L Make: P	Scale 1:1 A4
PA-CF or PETG, 100% infill (steel	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 20/50

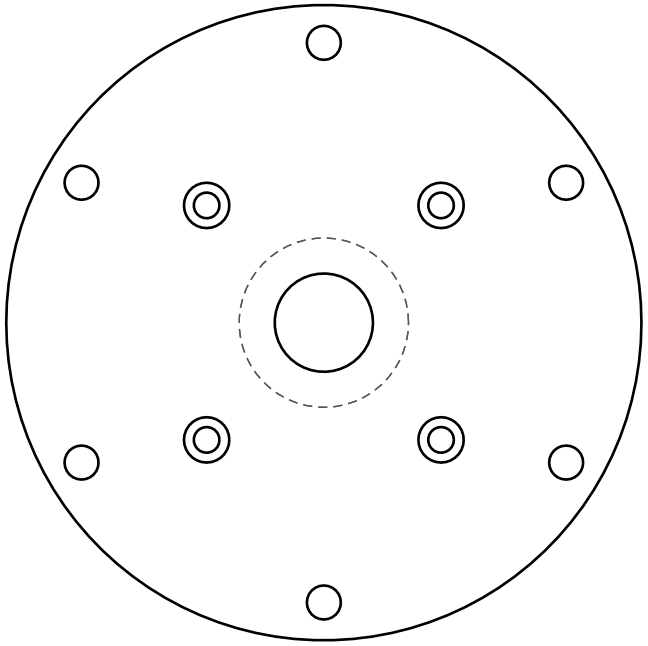
Planet carrier, config J2_LONG. Overall size 46.0 x 46.0 x 44.8 mm. Full dimensions: native SLDPRT / STEP model.



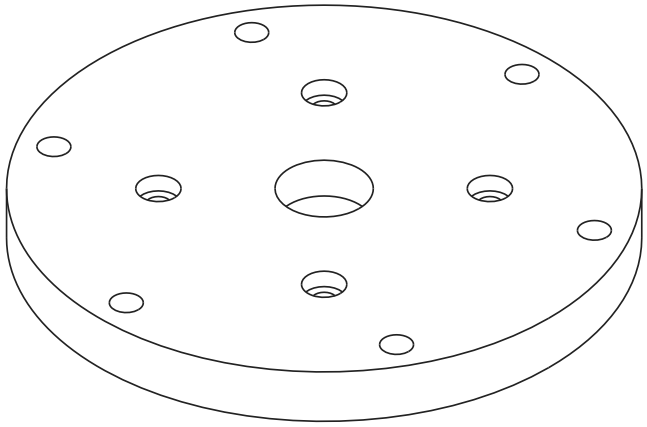
FRONT



LEFT



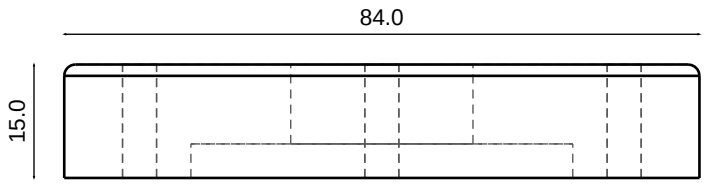
TOP (plan)



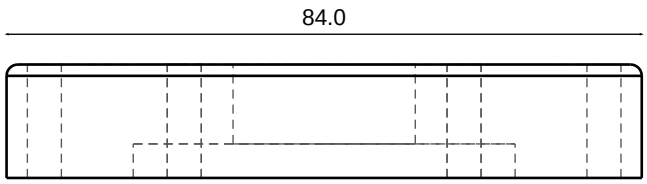
FIRST-ANGLE PROJECTION (ISO 128-30)

GBX_INPUT_HOUSING	
Part no. G05 Make: P	Scale 1:1 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 21/50

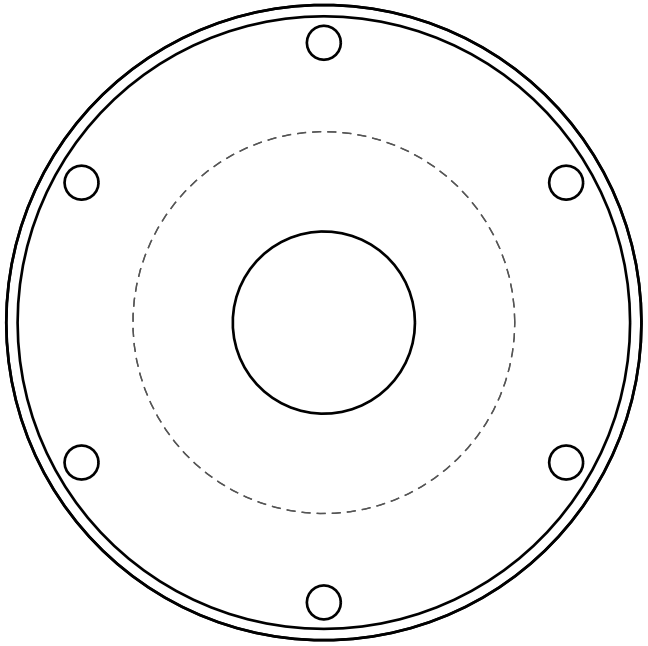
Motor-side housing plate. Overall size 84.0 x 84.0 x 8.0 mm. Full dimensions: native SLDPRT / STEP model.



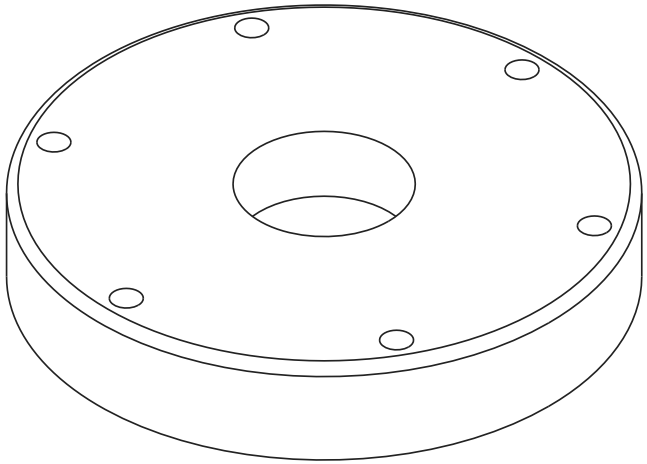
FRONT



LEFT



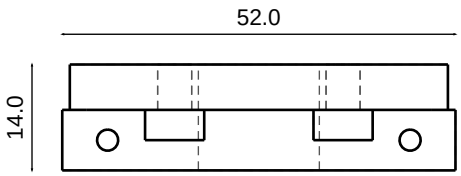
TOP (plan)



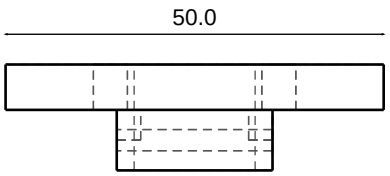
FIRST-ANGLE PROJECTION (ISO 128-30)

GBX_OUTPUT_COVER	
Part no. G06 Make: P	Scale 1:1 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 22/50

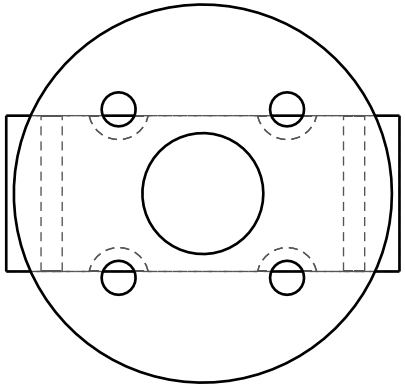
Output cover, 2x 6802 seats. Overall size 84.0 x 84.0 x 15.0 mm. Full dimensions: native SLDPRT / STEP model.



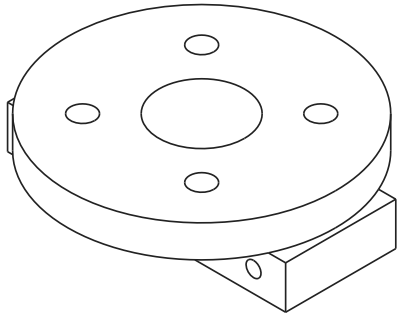
FRONT



LEFT



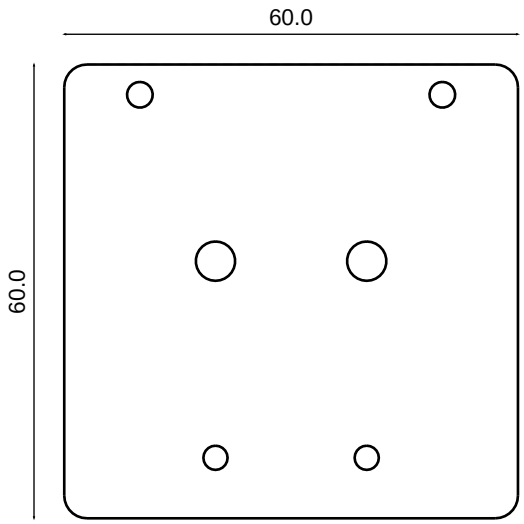
TOP (plan)



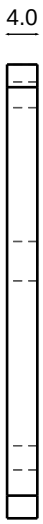
FIRST-ANGLE PROJECTION (ISO 128-30)

GRIPPER_MOUNT_PLATE	
Part no. P01 Make: P	Scale 1:1 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 23/50

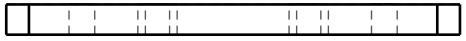
Adapter to tool flange. Overall size 52.0 x 50.0 x 14.0 mm. Full dimensions: native SLDPRT / STEP model.



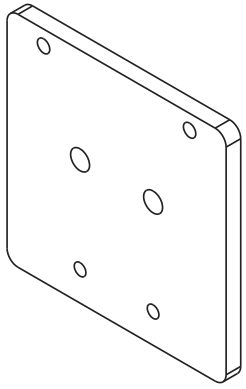
FRONT



LEFT



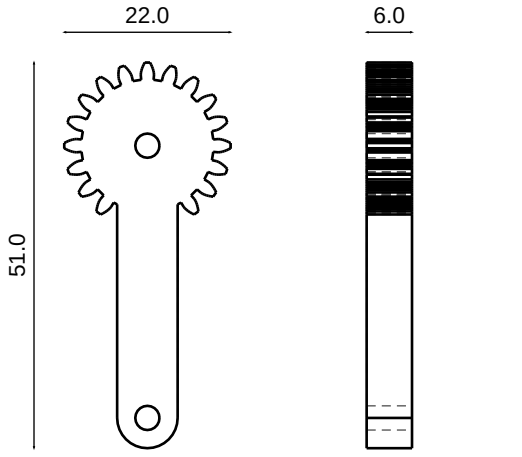
TOP (plan)



FIRST-ANGLE PROJECTION (ISO 128-30)

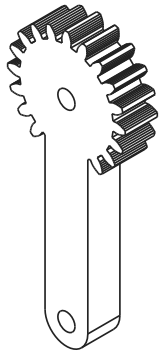
GRIPPER_HOUSING	
Part no. P02 Make: P	Scale 1:1 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 24/50

Side plate (front/back identical). Overall size 60.0 x 4.0 x 60.0 mm. Full dimensions: native SLDPRT / STEP model.





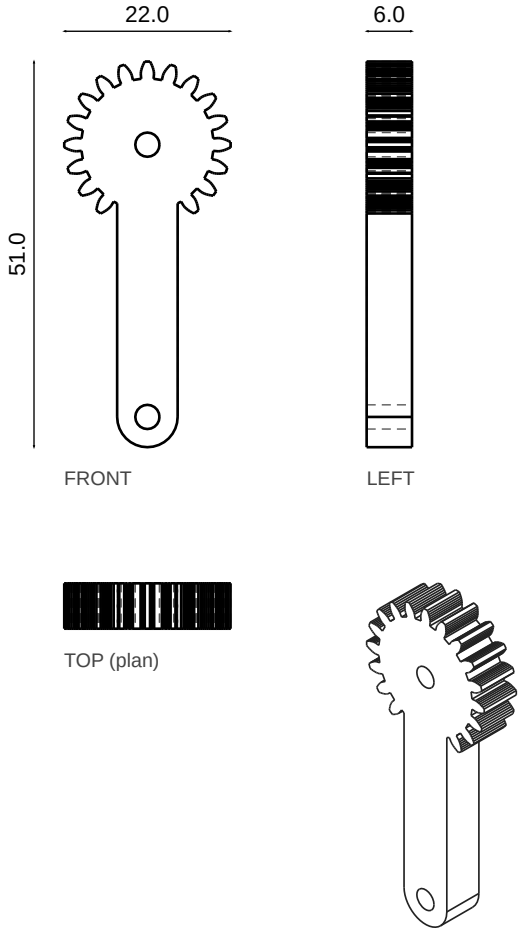
TOP (plan)



FIRST-ANGLE PROJECTION (ISO 128-30)

GRIPPER_DRIVE_GEAR_L	
Part no. P03 Make: P	Scale 1:1 A4
PETG, 100% infill	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 25/50

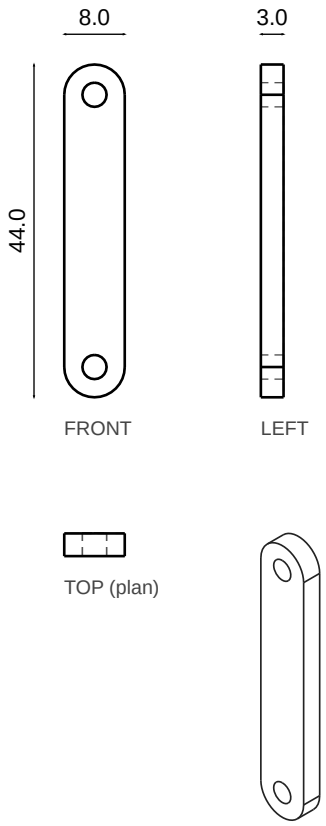
Drive gear m1 z20 + crank (servo side). Overall size 22.0 x 6.0 x 51.0 mm. Full dimensions: native SLDPRT / STEP model.



FIRST-ANGLE PROJECTION (ISO 128-30)

GRIPPER_DRIVE_GEAR_R	
Part no. P04 Make: P	Scale 1:1 A4
PETG, 100% infill	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 26/50

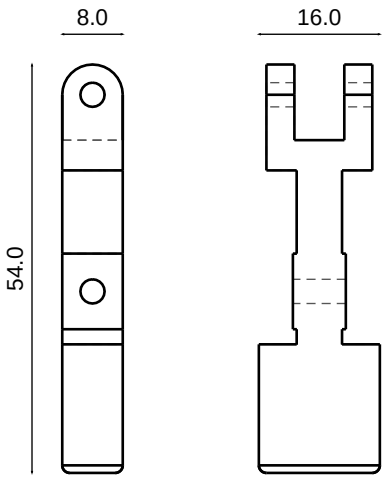
Driven gear m1 z20 + crank (assembled 9° phase). Overall size 22.0 x 6.0 x 51.0 mm. Full dimensions: native SLDPRT / STEP model.



FIRST-ANGLE PROJECTION (ISO 128-30)

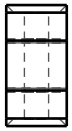
GRIPPER_LINK	
Part no. P05 Make: P	Scale 1:1 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 27/50

Parallel idler link 36 mm. Overall size 8.0 x 3.0 x 44.0 mm. Full dimensions: native SLDPRT / STEP model.

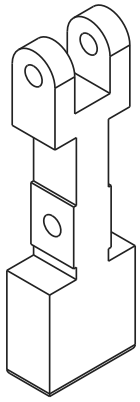


FRONT

LEFT



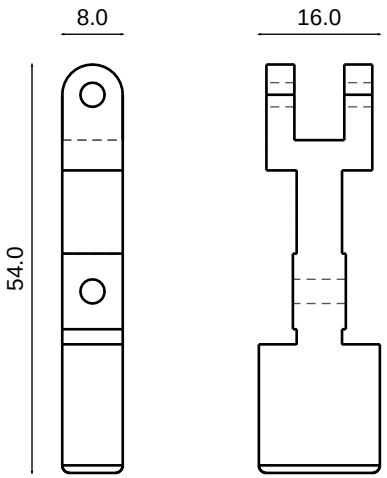
TOP (plan)



FIRST-ANGLE PROJECTION (ISO 128-30)

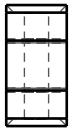
GRIPPER_FINGER_LEFT	
Part no. P06 Make: P	Scale 1:1 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 28/50

Left finger. Overall size 8.0 x 16.0 x 54.0 mm. Full dimensions: native SLDPRT / STEP model.

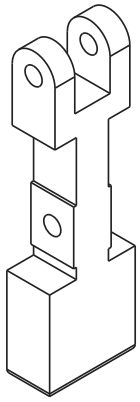


FRONT

LEFT



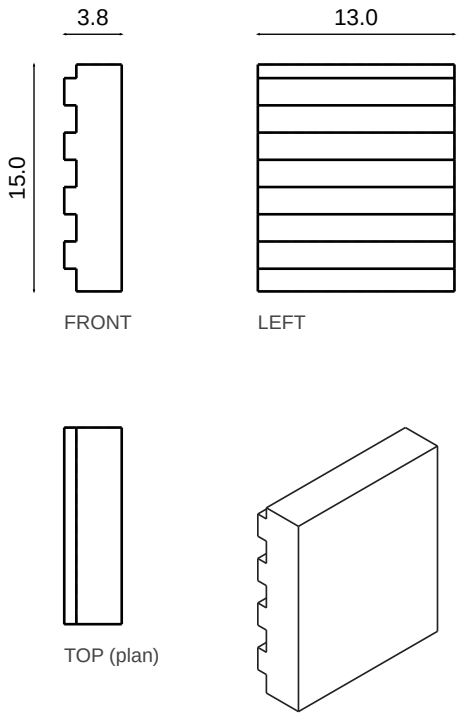
TOP (plan)



FIRST-ANGLE PROJECTION (ISO 128-30)

GRIPPER_FINGER_RIGHT	
Part no. P07 Make: P	Scale 1:1 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 29/50

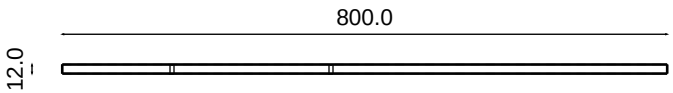
Right finger. Overall size 8.0 x 16.0 x 54.0 mm. Full dimensions: native SLDPRT / STEP model.



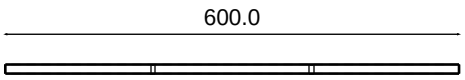
FIRST-ANGLE PROJECTION (ISO 128-30)

GRIPPER_PAD	
Part no. P08 Make: P	Scale 2:1 A4
TPU 95A	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 30/50

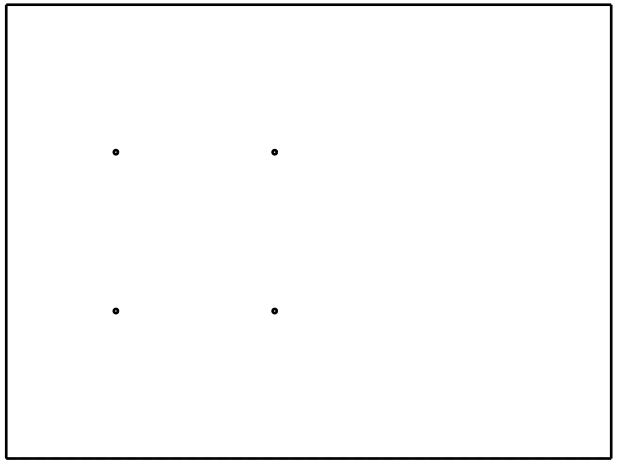
Ribbed pad (same part both sides, rotated 180 deg). Overall size 3.8 x 13.0 x 15.0 mm. Full dimensions: native SLDPRT / STEP model.



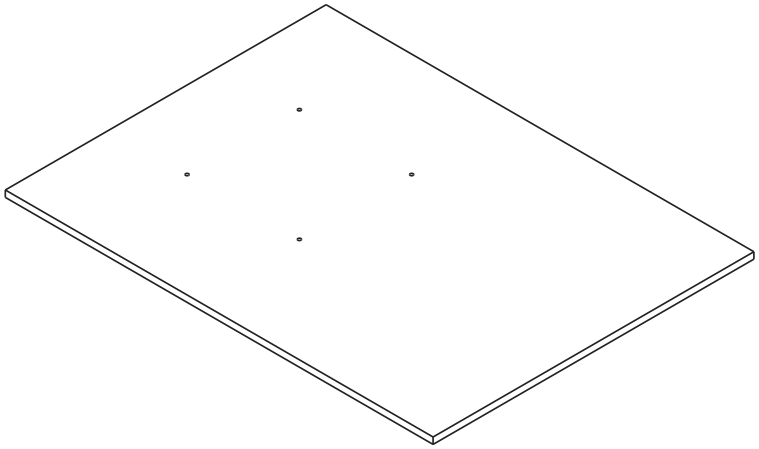
FRONT



LEFT



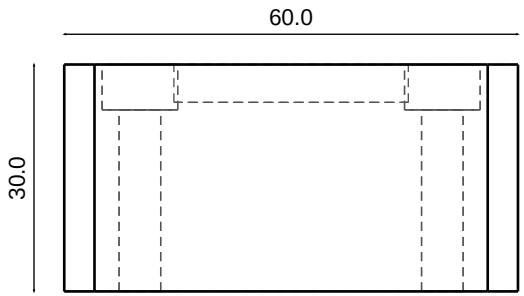
TOP (plan)



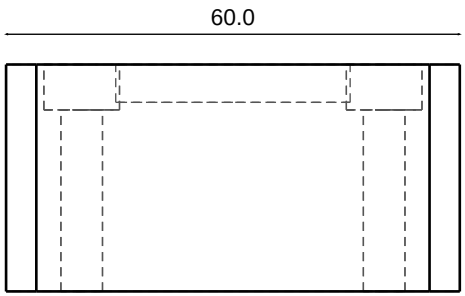
FIRST-ANGLE PROJECTION (ISO 128-30)

MACHINE_BASE_PLATE	
Part no. S01 Make: M	Scale 1:10 A4
Al 6061 or 12 mm birch ply	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 31/50

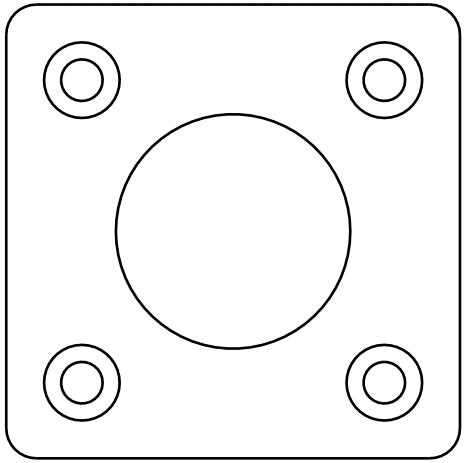
Cell base plate 800x600x12. Overall size 800.0 x 600.0 x 12.0 mm. Full dimensions: native SLDPRT / STEP model.



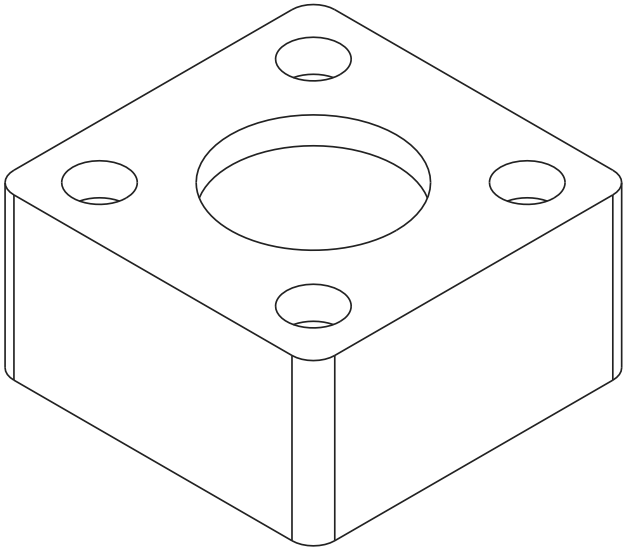
FRONT



LEFT



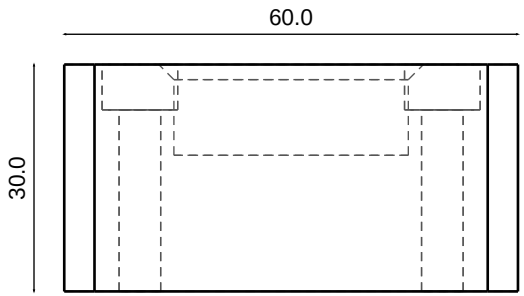
TOP (plan)



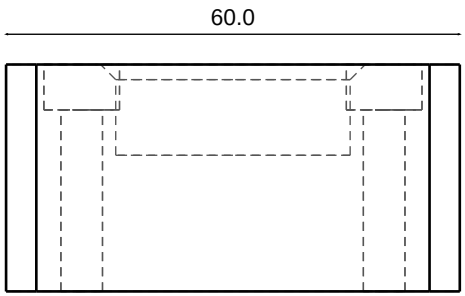
FIRST-ANGLE PROJECTION (ISO 128-30)

FIXTURE_PICK_BLOCK	
Part no. S02 Make: P	Scale 1:1 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 32/50

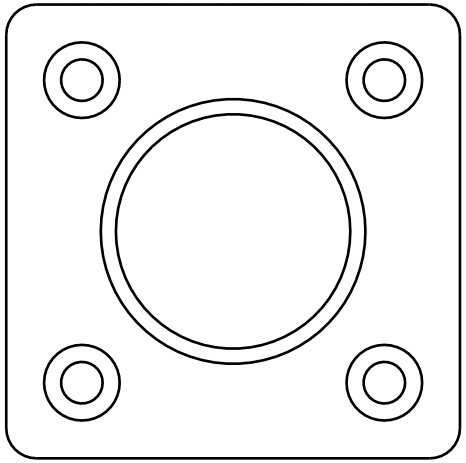
Pick fixture with 5 mm seat. Overall size 60.0 x 60.0 x 30.0 mm. Full dimensions: native SLDPRT / STEP model.



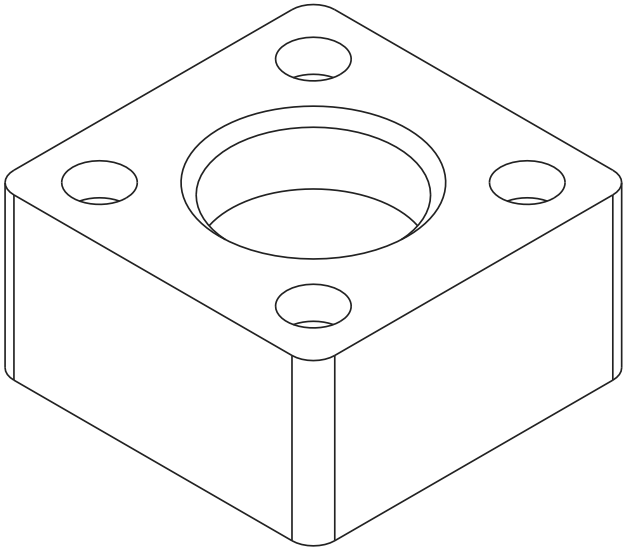
FRONT



LEFT



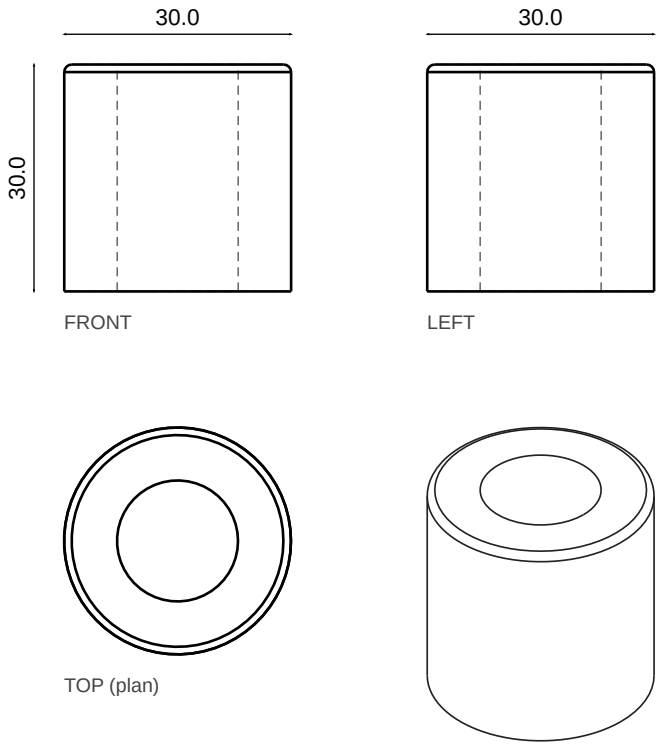
TOP (plan)



FIRST-ANGLE PROJECTION (ISO 128-30)

FIXTURE_PLACE_BLOCK	
Part no. S03 Make: P	Scale 1:1 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 33/50

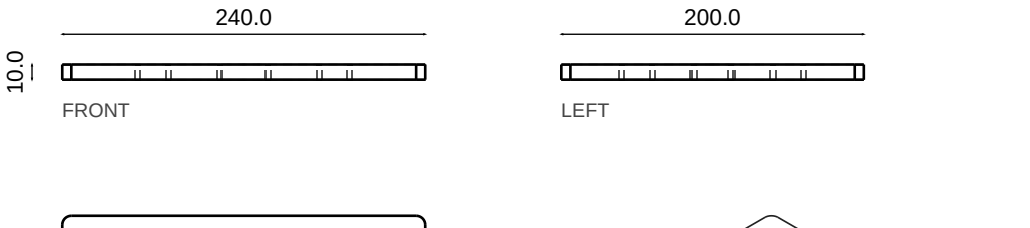
Place fixture with Ø31x12 hole. Overall size 60.0 x 60.0 x 30.0 mm. Full dimensions: native SLDPRT / STEP model.

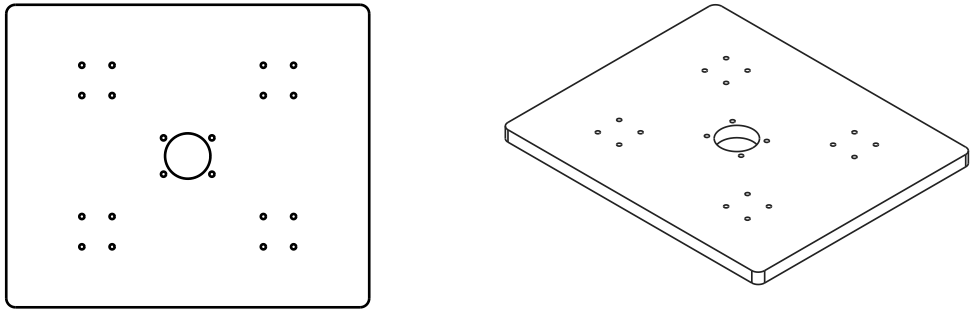


FIRST-ANGLE PROJECTION (ISO 128-30)

WORKPIECE_CYLINDER	
Part no. S04 Make: P	Scale 1:1 A4
PLA	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 34/50

Workpiece Ø30x30. Overall size 30.0 x 30.0 x 30.0 mm. Full dimensions: native SLDPRT / STEP model.



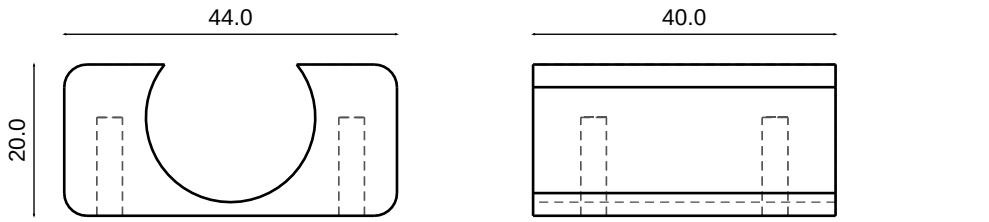


TOP (plan)

FIRST-ANGLE PROJECTION (ISO 128-30)

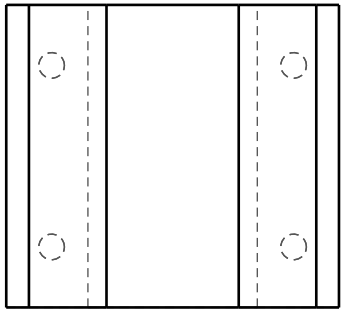
LINEAR_CARRIAGE_PLATE	
Part no. B04 Make: M	Scale 1:5 A4
Al 6061-T6, 10 mm plate	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 35/50

Carriage mounting plate (robot base bolts here). Overall size 240.0 x 200.0 x 10.0 mm. Full dimensions: native SLDPRT / STEP model.

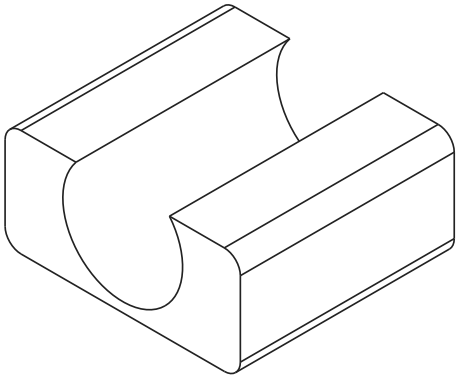


FRONT

LEFT



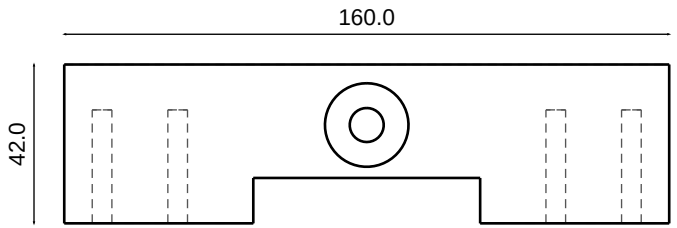
TOP (plan)



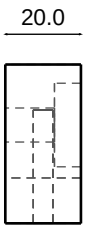
FIRST-ANGLE PROJECTION (ISO 128-30)

LINEAR_CARRIAGE	
Part no. B05 Make: P	Scale 1:1 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 36/50

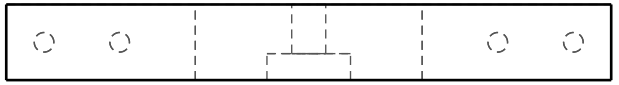
Moving carriage / T8 nut housing. Overall size 44.0 x 40.0 x 20.0 mm. Full dimensions: native SLDPRT / STEP model.



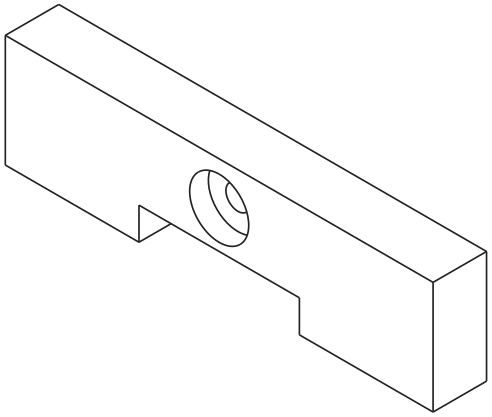
FRONT



LEFT



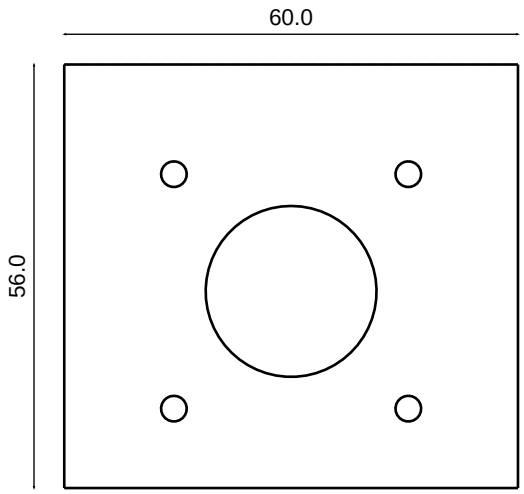
TOP (plan)



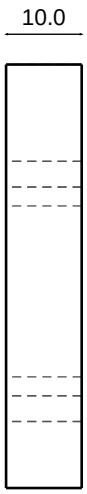
FIRST-ANGLE PROJECTION (ISO 128-30)

LINEAR_END_SUPPORT	
Part no. B08 Make: P	Scale 1:2 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 37/50

End support with 608 bearing seat. Overall size 160.0 x 20.0 x 42.0 mm. Full dimensions: native SLDPRT / STEP model.



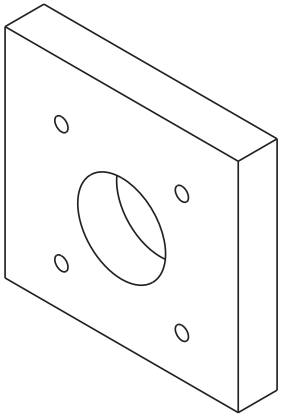
FRONT



LEFT



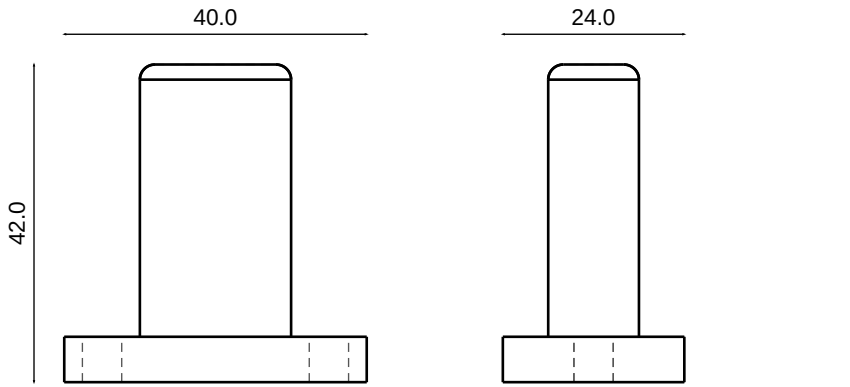
TOP (plan)

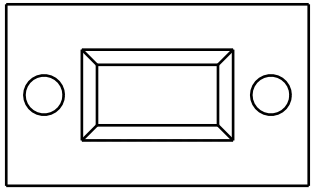


FIRST-ANGLE PROJECTION (ISO 128-30)

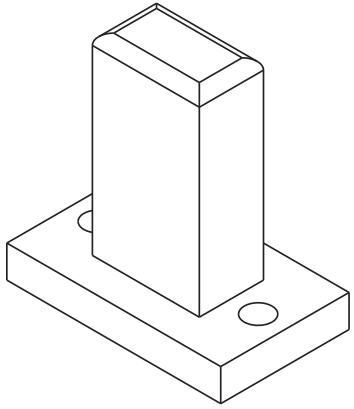
LINEAR_MOTOR_BRACKET	
Part no. B09 Make: P	Scale 1:1 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 38/50

Linear-axis motor bracket. Overall size 60.0 x 10.0 x 56.0 mm. Full dimensions: native SLDPRT / STEP model.





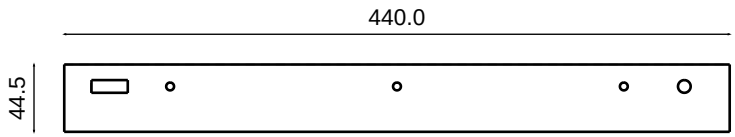
TOP (plan)



FIRST-ANGLE PROJECTION (ISO 128-30)

END_STOP	
Part no. B11 Make: P	Scale 1:1 A4
PETG / TPU	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 39/50

Mechanical end stop post. Overall size 40.0 x 24.0 x 42.0 mm. Full dimensions: native SLDPRT / STEP model.

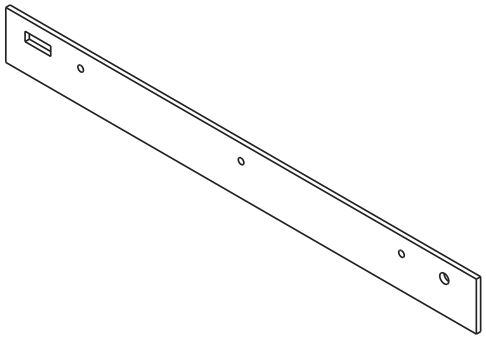


FRONT

LEFT



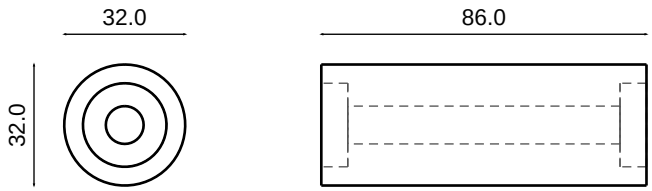
TOP (plan)

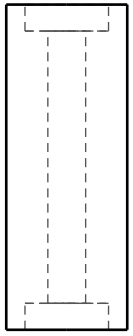


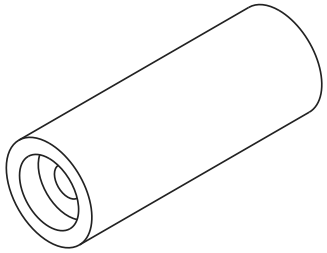
FIRST-ANGLE PROJECTION (ISO 128-30)

CONVEYOR_SIDE_PLATE	
Part no. C01 Make: M	Scale 1:5 A4
Al 5052, 4 mm	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 40/50

Conveyor side plate. Overall size 440.0 x 4.0 x 44.5 mm. Full dimensions: native SLDPRT / STEP model.



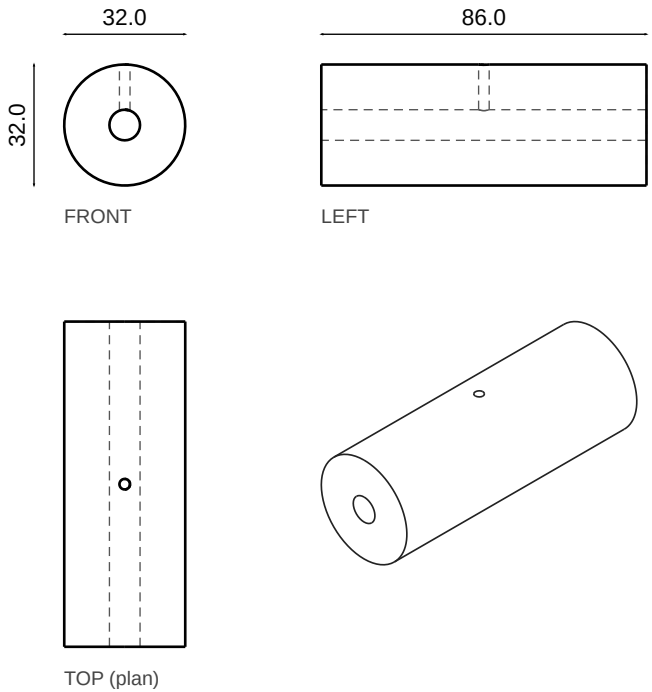

TOP (plan)



FIRST-ANGLE PROJECTION (ISO 128-30)

CONVEYOR_ROLLER	
Part no. C03 Make: P	Scale 1:2 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 41/50

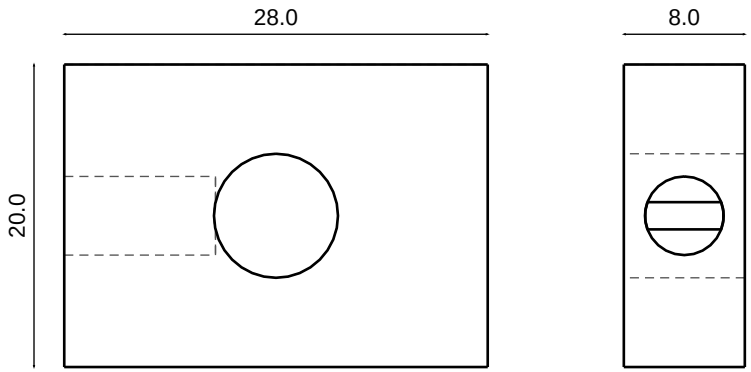
Idler roller with 2x 608 seats. Overall size 32.0 x 86.0 x 32.0 mm. Full dimensions: native SLDPRT / STEP model.

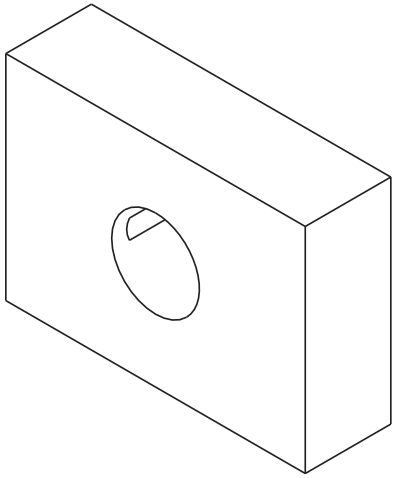


FIRST-ANGLE PROJECTION (ISO 128-30)

CONVEYOR_ROLLER_DRIVE	
Part no. C04 Make: P	Scale 1:2 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 42/50

Drive roller, pinned to shaft. Overall size 32.0 x 86.0 x 32.0 mm. Full dimensions: native SLDPRT / STEP model.

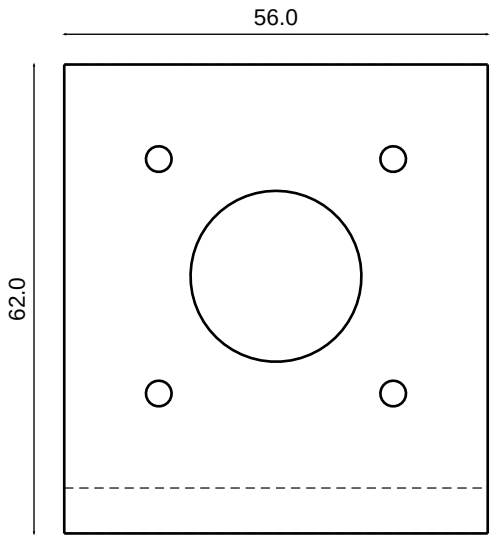




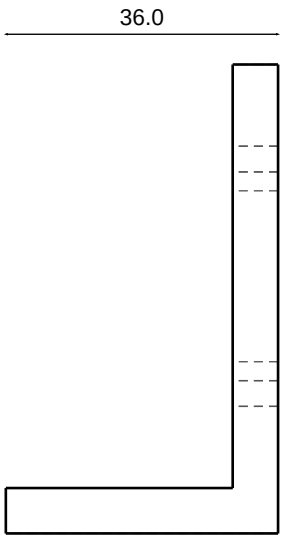
FIRST-ANGLE PROJECTION (ISO 128-30)

CONVEYOR_TENSIONER	
Part no. C07 Make: P	Scale 2:1 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 43/50

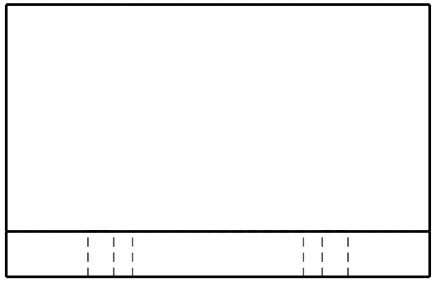
Sliding belt tensioner block. Overall size 28.0 x 8.0 x 20.0 mm. Full dimensions: native SLDPRT / STEP model.



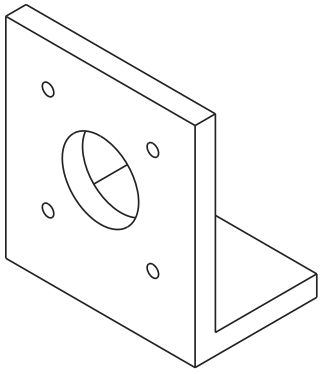
FRONT



LEFT



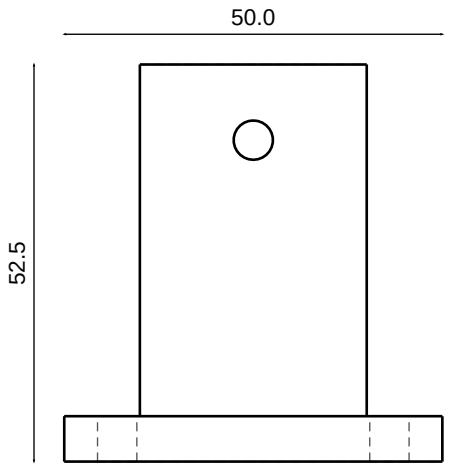
TOP (plan)



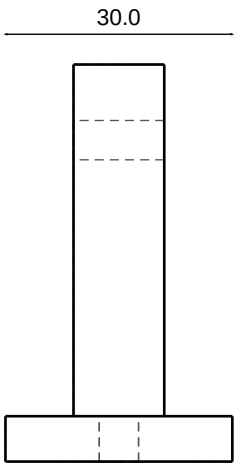
FIRST-ANGLE PROJECTION (ISO 128-30)

CONVEYOR_MOTOR_BRACKET	
Part no. C08 Make: P	Scale 1:1 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 44/50

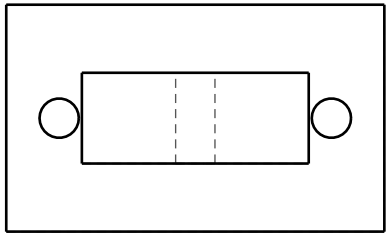
Conveyor motor bracket. Overall size 56.0 x 36.0 x 62.0 mm. Full dimensions: native SLDPRT / STEP model.



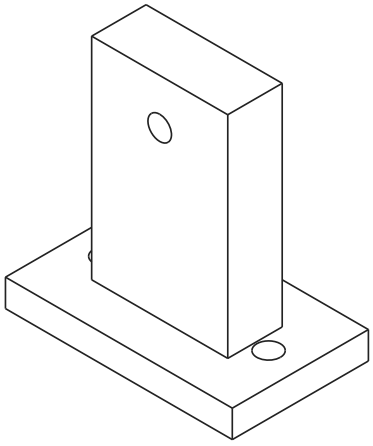
FRONT



LEFT



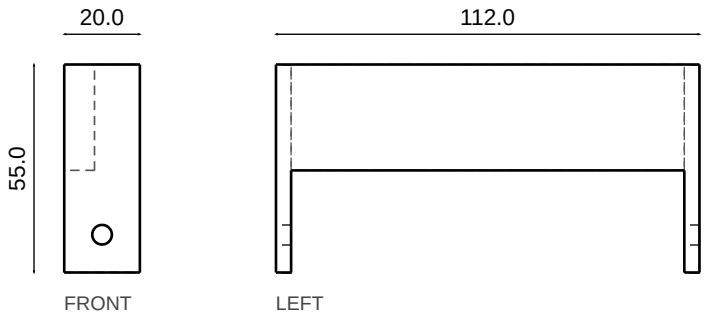
TOP (plan)

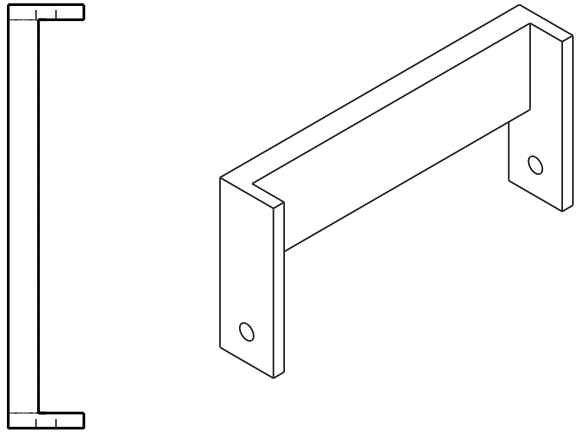


FIRST-ANGLE PROJECTION (ISO 128-30)

CONVEYOR_LEG	
Part no. C10 Make: P	Scale 1:1 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 45/50

Conveyor support leg. Overall size 50.0 x 30.0 x 52.5 mm. Full dimensions: native SLDPRT / STEP model.

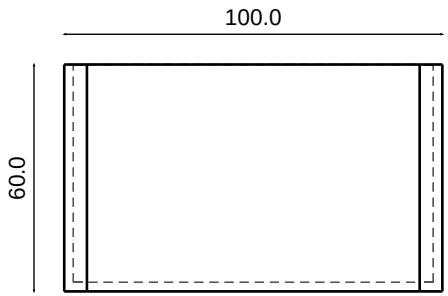




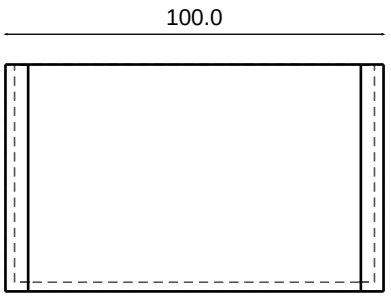
FIRST-ANGLE PROJECTION (ISO 128-30)

CONVEYOR_STOP	
Part no. C11 Make: P	Scale 1:2 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 46/50

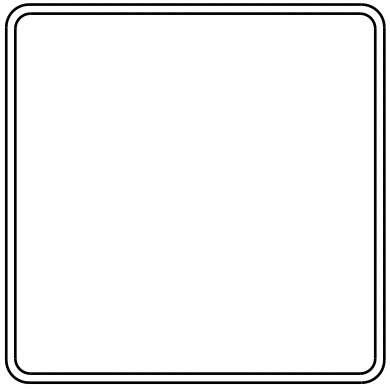
Pickup stop at belt end. Overall size 20.0 x 112.0 x 55.0 mm. Full dimensions: native SLDPRT / STEP model.



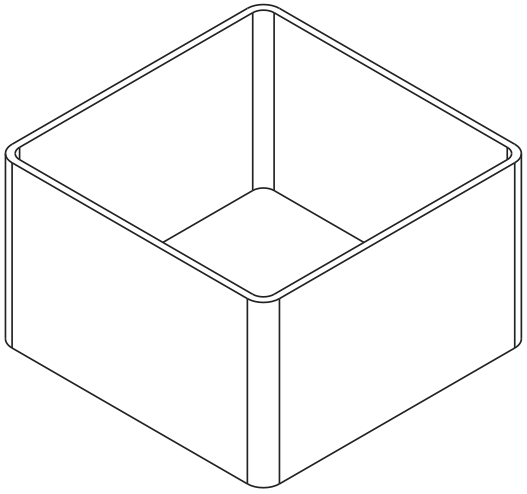
FRONT



LEFT



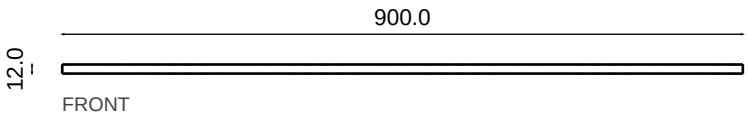
TOP (plan)



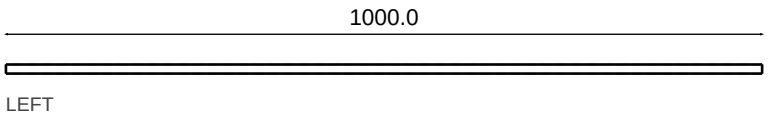
FIRST-ANGLE PROJECTION (ISO 128-30)

SORT_BIN	
Part no. S11 Make: P	Scale 1:2 A4
PETG	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 47/50

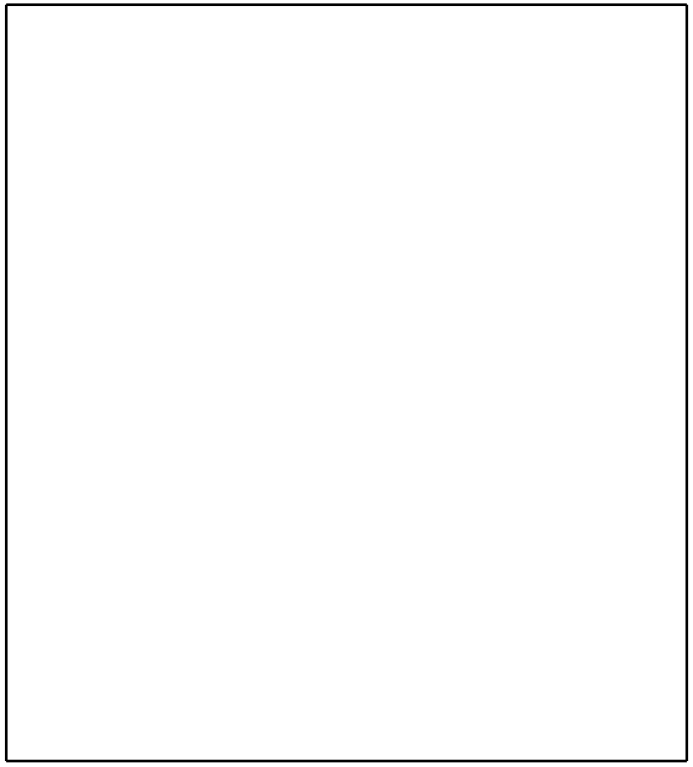
Sorting bin 100x100x60. Overall size 100.0 x 100.0 x 60.0 mm. Full dimensions: native SLDPRT / STEP model.



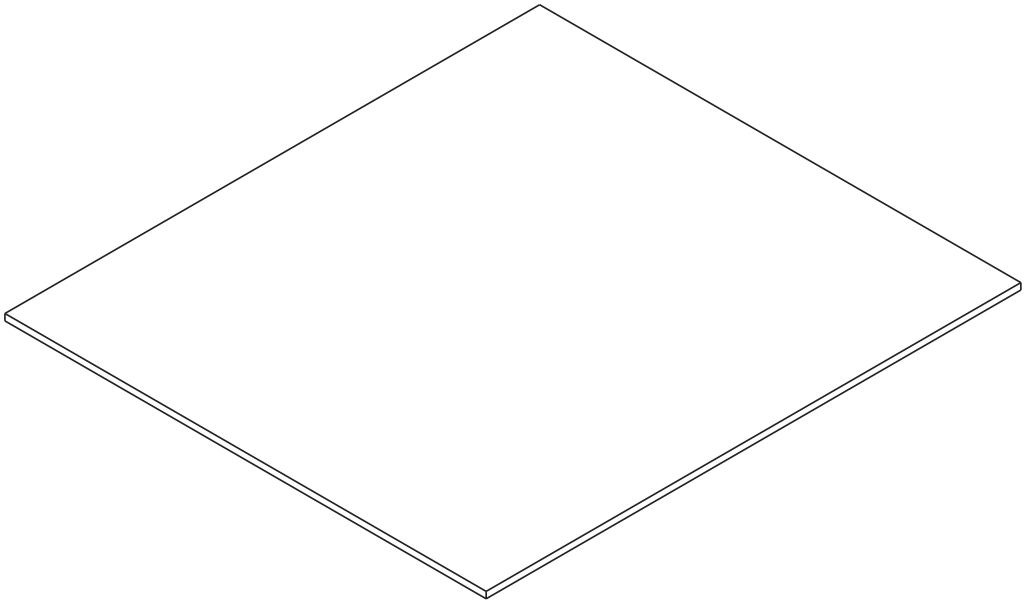
FRONT



LEFT



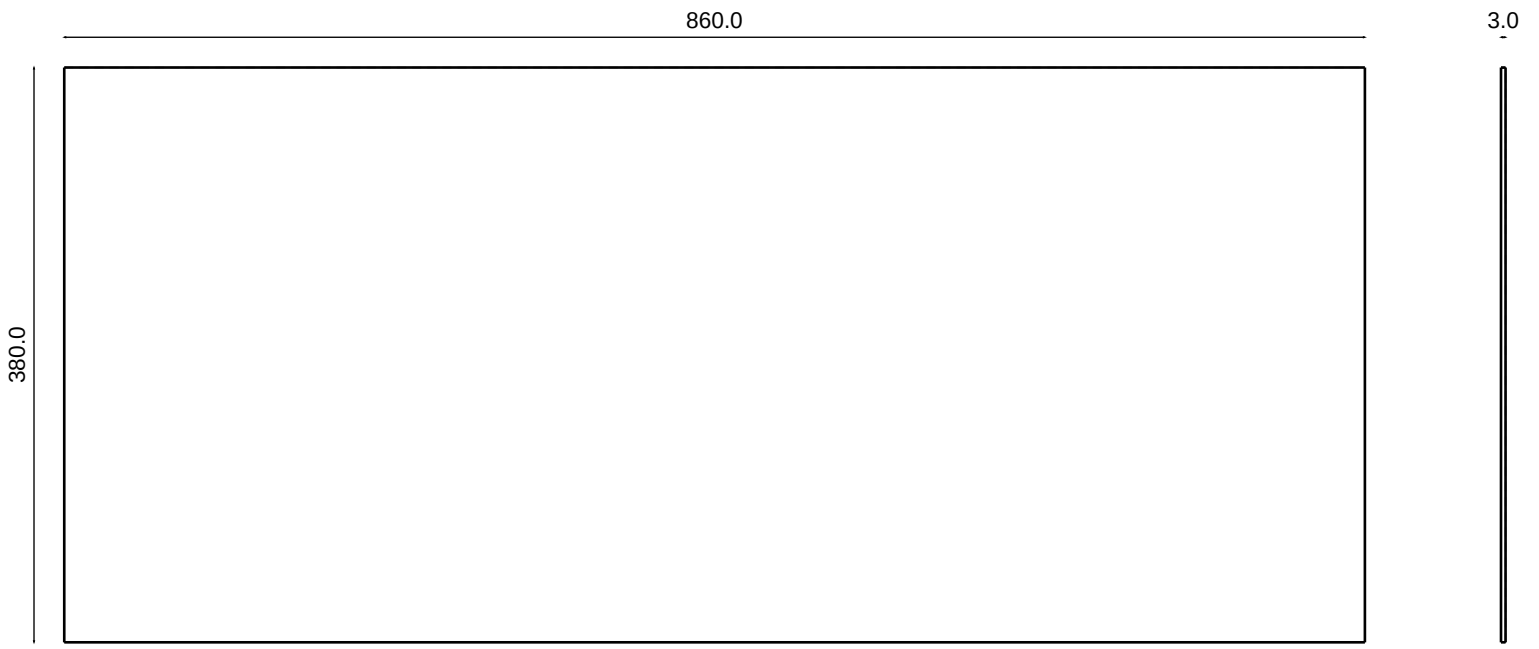
TOP (plan)



FIRST-ANGLE PROJECTION (ISO 128-30)

MACHINE_BASE_PLATE_B	
Part no. S10 Make: M	Scale 1:10 A4
Al 6061 or 18 mm ply	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 48/50

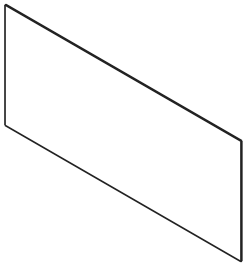
Cell base plate 900x1000 (full layout). Overall size 900.0 x 1000.0 x 12.0 mm. Full dimensions: native SLDPRT / STEP model.



FRONT

LEFT

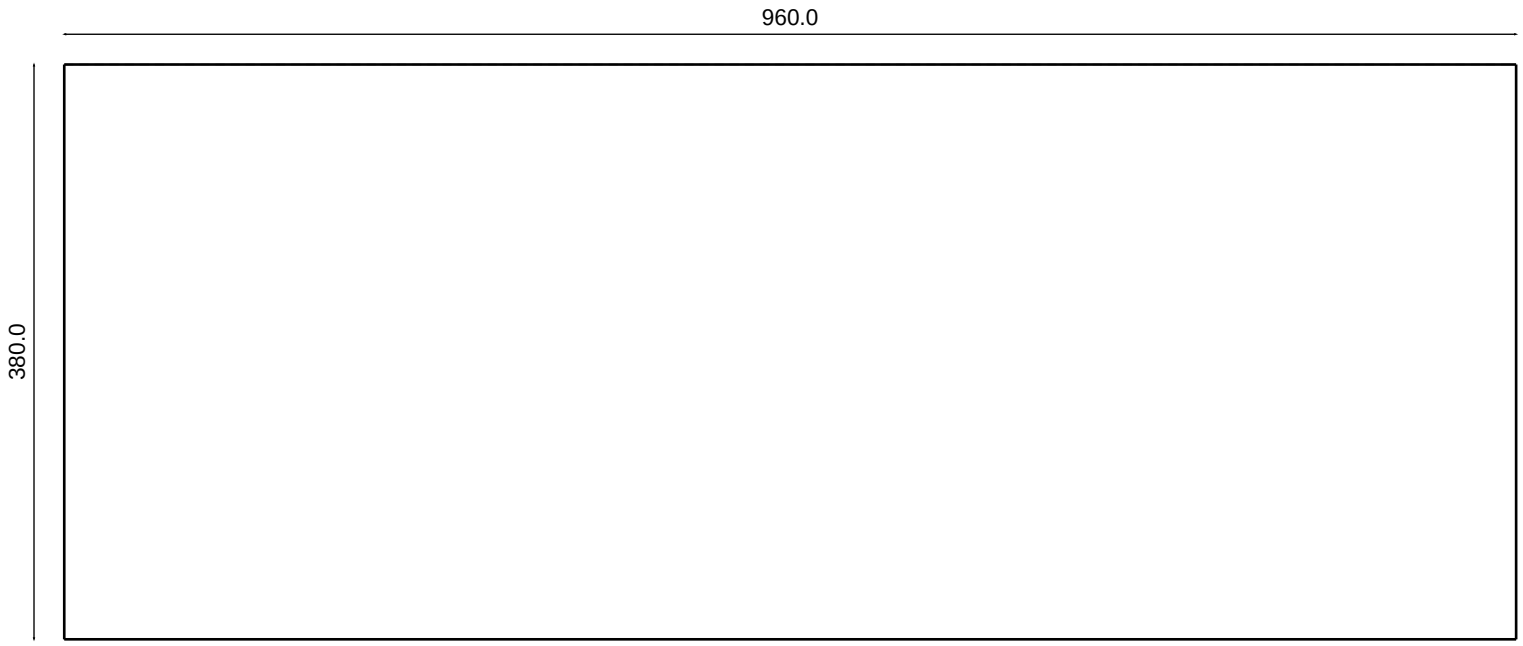
TOP (plan)



FIRST-ANGLE PROJECTION (ISO 128-30)

GUARD_PANEL_X	
Part no. S13 Make: M	Scale 1:5 A4
Polycarbonate 3 mm	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 49/50

Guard panel, long side. Overall size 860.0 x 3.0 x 380.0 mm. Full dimensions: native SLDPRT / STEP model.

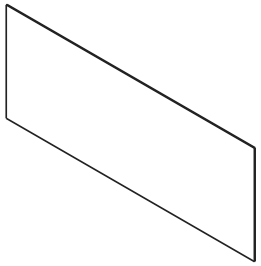


FRONT



LEFT

TOP (plan)



FIRST-ANGLE PROJECTION (ISO 128-30)

GUARD_PANEL_Y	
Part no. S14 Make: M	Scale 1:5 A4
Polycarbonate 3 mm	Units mm ISO 2768-m
PRELIMINARY - dims estimated	2026-11-01 sheet 50/50

Guard panel, short side. Overall size 960.0 x 3.0 x 380.0 mm. Full dimensions: native SLDPRT / STEP model.